



10. The volume and surface area of a family of regular solid shapes are related by the formula $V = kA^{1.5}$, where V is given in cm^3 and A in cm^2 .

(a) For one such shape, $A = 81$ and $V = 243$. Find k .

(b) Hence determine the surface area of a shape

with volume $\frac{64}{3} \text{ cm}^3$.

[4 marks]



11. Prove that 2^{350} is larger than 5^{150} .

[5 marks]



12. Given that there is more than one solution value of x to the equation $4^{ax} = b \times 8^x$, find all possible values of a and b .

[5 marks]

2B Exponential functions

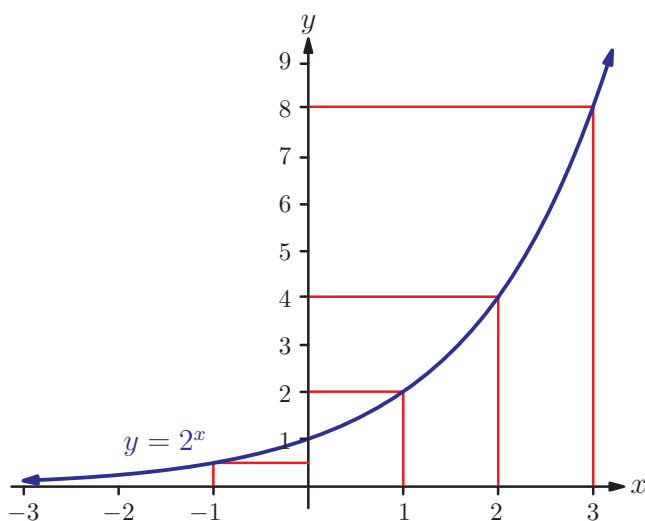
In most of the equations that you have met so far, the unknown appears as the base, for example $x^3 = 27$.

In an exponential function, the unknown appears in the exponent, leading to a fundamentally different type of function.

The general form of a simple exponential function is $f(x) = a^x$.

We will only consider situations when a is positive, because otherwise some exponents cannot be easily defined (for example, we cannot square root a negative number).

Here is the graph of $y = 2^x$:

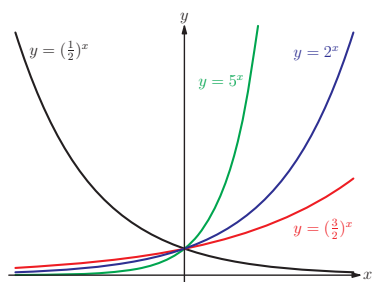


What is $(-1)^{\frac{1}{2}}$?

What about $((-1)^2)^{\frac{1}{4}}$?

What about $(-1)^{\frac{2}{4}}$?

Not all mathematics is unambiguous!



You may observe that the blue line is a reflection of the black line. You will see why this is the case in chapter 6.

For very large positive values of x , the y -value approaches infinity. For very large negative values of x , the y -value approaches (but never reaches) zero.

A line that a function gets increasingly close to but never reaches is called an **asymptote**. In this case we would say that the x -axis is an asymptote to the graph.

If we look at the graphs of other exponential functions with different bases we can begin to make some generalisations.

For all the graphs $y = a^x$:

- The y -intercept is always $(0, 1)$ because $a^0 = 1$.
- The graph of the function lies entirely above the x -axis since $a^x > 0$ for all values of x .
- The x -axis is an asymptote.
- If $a > 1$, then as x increases, so does $f(x)$. This is called a **positive exponential**.
- If $0 < a < 1$, then as x increases, $f(x)$ decreases. This is called a **negative exponential**.

Many mathematical models use the characteristics of exponential functions: as time (t) increases by a fixed value, the value we are interested in (N) will change by a fixed **factor**, called the **growth factor**. Exponential functions can therefore be used to model many physical, financial and biological forms of **exponential growth** (positive exponential models) and **exponential decay** (negative exponential models).

To model more complex situations we may need to add more constants to our exponential equation.

We can use a function of the form $N = Ba^{\left(\frac{t}{k}\right)}$.

We can interpret the constants in the following way:

- When $t = 0$, $N = B$ so B is the **initial value** of N .
- When $t = k$, $N = Ba$ so k is the time taken for N to increase by a factor of a .
- If $k = 1$ then a is the growth factor.
- As long as k is positive when $a > 1$ the function models exponential growth.
- When $0 < a < 1$ the function models exponential decay.
- When modelling exponential decay there might sometimes be a **background level**. This means the asymptote is not $N = 0$.

We can change the asymptote to $N = c$ by adding on a constant.

The new function is then $N = Ba^{\left(\frac{t}{k}\right)} + c$.

B represents how much N starts above the background level, so the initial value is $B + c$.

KEY POINT 2.6

If $N = Ba^{\left(\frac{t}{k}\right)} + c$:

- The background level (and the asymptote) is c .
- The initial value is $B + c$.
- k is the time taken for the difference between N and the background level to increase by a factor of a .
- If $a > 1$ this models exponential growth.
- If $0 < a < 1$ this models exponential decay.

In many applications we are given certain information and need to find the constants in the model.

EXAM HINT

See Calculator skills sheet 2 on the CD-ROM to find out how to sketch graphs on your calculator. As asymptotes are not a part of the graph, your calculator might not show them, though you can guess approximately where they are by looking at large values of x . This is why it is important to know how to find asymptotes directly from the equation.



Worked example 2.3

A population of bacteria in a culture medium doubles in size every 15 minutes.

- Write down a model for N , the number of bacteria in terms of time, t , in hours.
- If there are 1000 bacterial cells at 08:00, how many cells are there at
 - 08:15?
 - 09:24?

There is a constant increasing factor so use an exponential growth model

Every time t increases by 0.25, N doubles

All details are relative to a start time 08:00, so set $t = 0$ at that time

Remember to convert minutes to hours

- Let N be the number of cells at time t hours

$$N = Ba^{\left(\frac{t}{k}\right)}$$

Doubles every quarter hour

$$a = 2, k = 0.25$$

$$\therefore N = B \times 2^{4t}$$

- When $t = 0$, $N = 1000 = B$
 $N = 1000 \times 2^{4t}$

- When $t = 0.25$, $N = 2000$ cells
- When $t = 1.4$, $N = 48\,503$ cells

You may also be given a model and have to interpret it. Unfortunately, it might not be given in exactly the same form as in Key point 2.6 $\left(N = Ba^{\frac{t}{k}}\right)$, so you will have to use the rules of exponents to rewrite it in the correct form.

Worked example 2.4

The temperature in degrees (θ) of a cup of coffee a time t minutes after it was made is modelled using the function:

$$\theta = 70 \times 3^{-\beta t} + 22, \beta > 0$$

- Show that $\theta - 22$ follows an exponential decay.
- What was the initial temperature of the coffee?
- If the coffee is left for a very long time, what temperature does the model predict it will reach?
- Find, in terms of β , how long it takes for the temperature difference between the coffee and the room to fall by a factor of 9.
- Coffee made at the same time is put into a thermos flask which is much better insulated than the cup. State with a reason whether β would increase or decrease.

Use rules of exponents to rewrite the exponent in the form $N = Ba^{\left(\frac{t}{k}\right)}$

$$\begin{aligned} \text{(a) } \theta - 22 &= 70 \times 3^{-\beta t} \\ &= 70 \times (3^{-1})^{\beta t} \\ &= 70 \times \left(\frac{1}{3}\right)^{\frac{t}{\beta^{-1}}} \end{aligned}$$

which is exponential decay since the base is between 0 and 1 and $\beta > 0$

Initial means $t = 0$

$$\begin{aligned} \text{(b) When } t = 0, \theta &= 70 \times 3^0 + 22 \\ &= 92^\circ \end{aligned}$$

The asymptote is given by the background level

$$\text{(c) } 22 \text{ degrees}$$

Use algebra to rewrite the exponent in the form $N = Ba^{\frac{t}{k}}$

$$\text{(d) } \theta - 22 = 70 \left(\frac{1}{3}\right)^{\frac{t}{\beta^{-1}}}$$

So the time to fall by a factor of 3 is $\frac{1}{\beta}$

To fall by a factor of 9 is a fall by a factor of 3 followed by another fall by a factor of 3. Therefore the time to fall by a factor of 9 is $2 \times \frac{1}{\beta}$ minutes.

continued...

- (e) The time taken to fall by any given factor will be longer for a better insulated container so $\frac{1}{\beta}$ must be larger so β must be smaller.

When modelling exponential growth or decay, you may be given a percentage increase or decrease. This needs to be converted into a growth factor to be used in the exponential model.

Worked example 2.5

A car costs \$17 500. It then loses value at a rate of 18% each year.

- (a) Write a model for the value of the car (V) after n years in the form $V = ka^n$.
(b) Hence or otherwise find the value of the car after 20 years.

Find the growth factor

Use initial value information

Substitute for n

(a) The growth factor is $1 - \frac{18}{100} = 0.82$

When $n = 0$, $V = k = 17500$

$$V = 17500 \times 0.82^n$$

(b) After 20 years, $V = \$330.61$

Exercise 2B

1. Using your calculator, sketch the following functions for $-5 \leq x \leq 5$ and $0 \leq y \leq 10$.

Show all the axis intercepts and state the equation of the horizontal asymptote.

- | | |
|--|---------------------------------------|
| (a) (i) $y = 1.5^x$ | (ii) $y = 3^x$ |
| (b) (i) $y = 2 \times 3^x$ | (ii) $y = 6 \times 1.4^x$ |
| (c) (i) $y = \left(\frac{1}{2}\right)^x$ | (ii) $y = \left(\frac{2}{3}\right)^x$ |
| (d) (i) $y = 5 + 2^x$ | (ii) $y = 8 + 3^x$ |
| (e) (i) $y = 6 - 2^x$ | (ii) $y = 1 - 5^x$ |

2. An algal population on the surface of a pond grows by 10% every day, and the area it covers can be modelled by the equation $y = k \times 1.1^t$, where t is measured in days. At 09:00 on Tuesday it covered 10 m^2 . What area will it cover by 09:00 on Friday? [4 marks]

3. The air temperature $T^{\circ}\text{C}$ around a light bulb is given by equation

$$T = A + B \times 2^{-\frac{x}{k}}$$

where x is the distance from the surface of the light bulb in millimetres. The background temperature in the room is a constant 25°C , and the temperature on the surface of the light bulb is 125°C .

- (a) You find that the air temperature 3 mm from the surface of the bulb is only 75°C . Find the integer values of A , B and k .
- (b) Determine the air temperature 2 cm from the surface of the bulb.
- (c) Sketch a graph of air temperature against distance.

[10 marks]

4. A tree branch is observed to bend as the fruit growing on it increases in size. By estimating the mass of the developing fruit, and plotting the data over time, a student finds that the height in metres of the branch tip above the ground closely follows the graph of:

$$h = 2 - 0.2 \times 1.6^{0.2m}$$

where m is the estimated mass, in kilograms, of fruit on the branch.

- (a) Plot a graph of h against m .
- (b) What height above ground level is a branch without fruit?
- (c) The total mass of fruit on the branch at harvest was 7.5 kg. Find the height of the branch immediately prior to harvest.
- (d) The student wishes to estimate what mass of fruit would cause the branch tip to touch the ground. Why might his model not be suitable to assess this?

[10 marks]

5. (a) Sketch the graph of $y = 1 + 16^{1-x^2}$.

Label clearly the horizontal asymptote and maximum value.

- (b) Find all values of x for which $y = 3$.

[6 marks]

6. A bowl of soup is served at a temperature of 55°C in a room with a constant air temperature of 20°C . Every 5 minutes, the temperature difference between the soup and the room air decreases by 30%. Assuming the room air temperature is constant, at what temperature will the soup be seven minutes after serving?

[7 marks]

In this chapter you will learn:

- about the concepts of relations and functions, and how to distinguish between them
- the notation used to represent functions
- what happens when one function acts after another function
- how to reverse the effect of a function
- how to apply the ideas from this chapter to analyse functions formed by dividing one polynomial by another.

5 The theory of functions

Introductory problem

Think of any number. Add 3. Double your answer. Take away 6. Divide by the number you first thought of. Is your answer always prime? Why?

Doubling; adding five; finding the largest prime factor; these are all instructions which can be applied to different numbers in order to get a result. This idea is used a lot in mathematics, and its formal study leads to the concept of functions.

5A Relations, functions and graphs

This section introduces the concepts of a function and a relation and how to distinguish between them.

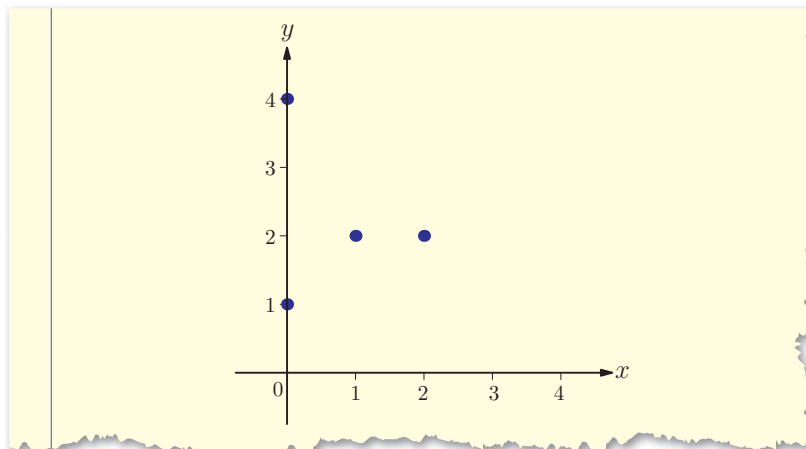
A **relation** can be written as a list of ordered pairs. An example of a relation might link a person to their age:

(John, 7)
(Patricia, 23)
(Joanne, 14)

Most of the relations we are going to use will be pairs of real numbers and we can illustrate these on a graph by plotting them as points. Conventionally we plot the first number on the x -axis and the second number on the y -axis.

Worked example 5.1

Plot the relation $(0, 1)$ $(0, 4)$ $(1, 2)$ $(2, 2)$ on a graph.



A relation can be more than just a few scattered points. We can have an infinite number of points, but these would be impossible to list. Instead we provide a rule that links the x -coordinate and the y -coordinate. Examples of such rules are $y = 3x$ or $y = \text{a factor of } x$.

For the relation $y = \text{a factor of } x$ you will realise that for some numbers there is more than one answer and so we cannot be sure what y is. This is a problem for mathematicians who like to work without ambiguity. A solution is to restrict their attention only to relations where for each x -value there is only one y -value. This type of relation is called a **function**. The easiest way to identify if a relation is a function is to look at a graph of the function, using a technique called the **vertical line test**.

KEY POINT 5.1

Vertical line test

If a relation is a function, any vertical line will cross its graph no more than once.

When we study inverse functions you will see that functions are further divided into two categories: one-to-one functions and many-to-one functions.

For a **one-to-one function**, as well as every x -value corresponding to only one y -value, every y -value corresponds to only one x -value.

Therefore, as well as the vertical line test a one-to-one function passes the **horizontal line test**.

Inverse functions are covered in Section E of this chapter.

KEY POINT 5.2

Horizontal line test

If a function is one-to-one then any horizontal line will cross the graph no more than once.

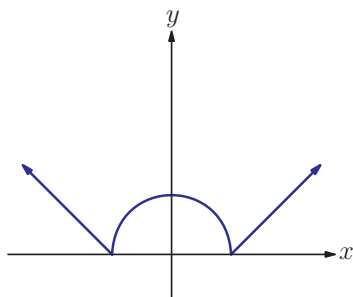
If a function does *not* pass the horizontal line test then there are some y -values which come from more than one x -value. This is a **many-to-one function**.

Worked example 5.2

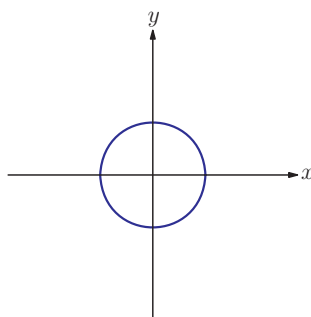
Which of the graphs below represent functions?

For those that are functions, classify them as one-to-one or many-to-one.

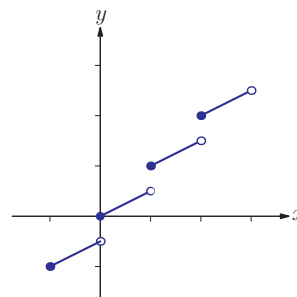
(a)



(b)



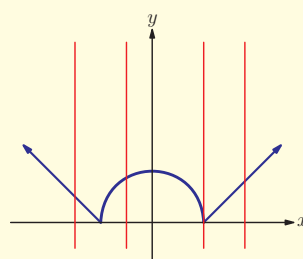
(c)



Note: The white circles mean that the end point is not included, and filled circles mean that it is.

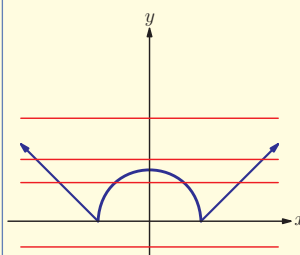
Apply the vertical line test to see whether the relation is a function

(a)



Any vertical line meets the graph at most once, therefore it is a function.

Apply the horizontal line test to see whether the function is one-to-one



continued...

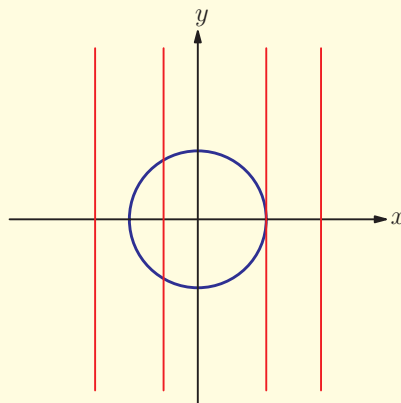
Apply the vertical line test to see whether the relation is a function

Apply the vertical line test to see whether the relation is a function. A vertical line through an open circle does not count as an intersection

Apply the horizontal line test to see whether the function is one-to-one

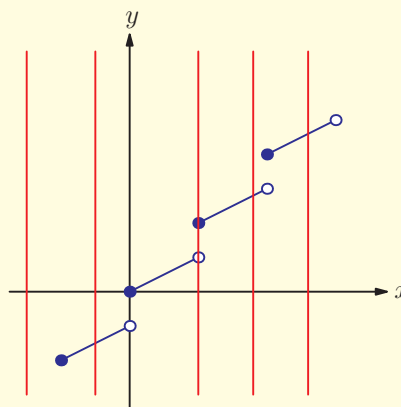
Some horizontal lines meet the graph at more than one point therefore it is a many-to-one function.

(b)

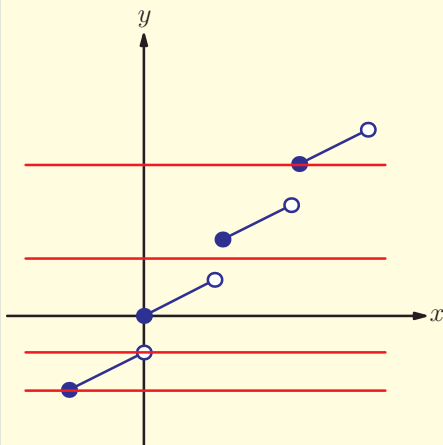


Some vertical lines meet more than once therefore it is not a function.

(c)



Any vertical line meets the graph at most once, therefore it is a function.



Any horizontal line meets the graph at most once, therefore it is a one-to-one function.

Often a relation is given by an equation linking x and y and we may not know how to draw its graph. In that case we need to look at the equation and ask whether there are any values of x for which we can find more than one value of y .

Worked example 5.3

Decide whether the following relations are functions:

- (a) $x + y^2 = 5$ (b) $x + y^3 = 9$

Think of an example where there is more than one y for each x

Can we prove that there is always just one value of y ? Try to find y for a given x

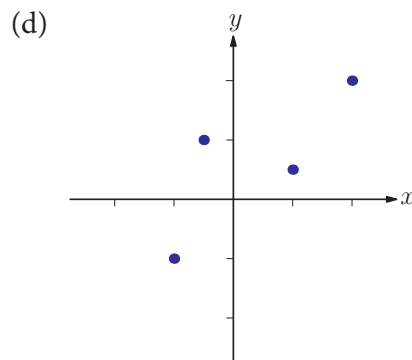
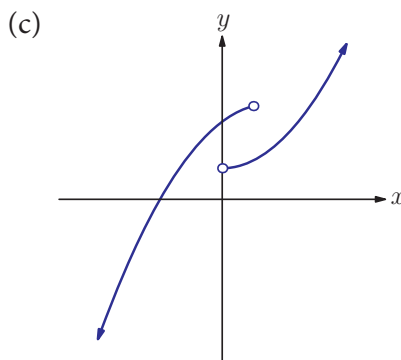
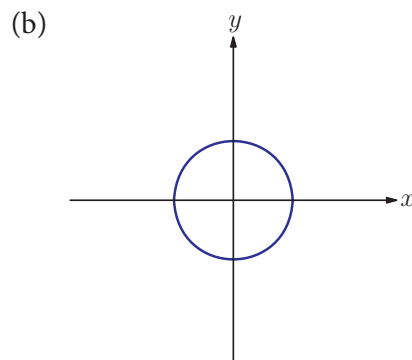
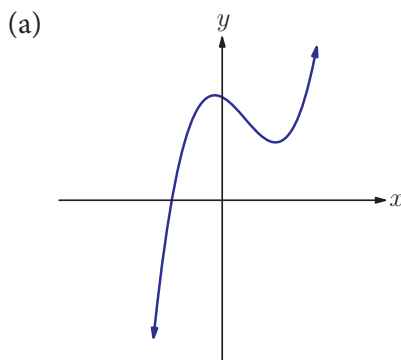
(a) For $x = 1$, y could be 2 or -2
So this is not a function.

$$\begin{aligned} \text{(b)} \quad x + y^3 &= 9 \\ \Leftrightarrow y^3 &= 9 - x \\ \Leftrightarrow y &= \sqrt[3]{9 - x} \end{aligned}$$

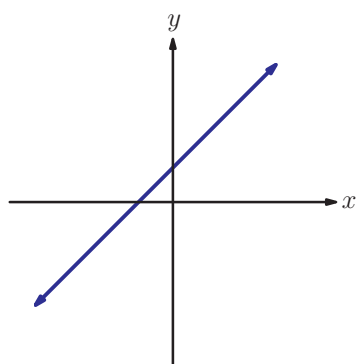
There is only one y for each x .
So this is a function.

Exercise 5A

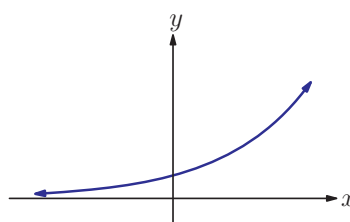
1. Classify these graphs as relations or functions. If you decide a graph is a function, state if it is one-to-one or many-to-one.



(e)



(f)



2. Decide if the following relations are functions:

(a) (i) $y = \sqrt{x}$

(ii) $y = 3x + 7$

(b) (i) $y = 5x^2 + 3x$

(ii) $y = 9 - x^2$

(c) (i) $y^2 = x$

(ii) $x^2 + y^2 = 1$

5B Function notation

In this section we will see how to describe functions using mathematical expressions. One way of thinking about functions is that they are rules for turning one number into another. If we have the rule 'add 3 to the input' and we want to call this rule ' f ', in function notation this is written as:

$$f: x \rightarrow x + 3$$

We say that the function f transforms x into $x + 3$.

An alternative way of writing this is

$$f(x) = x + 3.$$

Notice that although conventionally the letters f and x are often used, they are not particularly important and we can replace f with any name for the function and x with any input we liked. But whatever we call the input, f will do the same thing to it. We sometimes also call the input the **argument** of the function.

The output is called the **image** of the input.



Is $f(x)$ just a tool to label a rule or does it open up new techniques and new knowledge?

It may surprise you to learn that the latter is actually the case. Particularly in many applications of calculus, we do not need to know exactly what the rule is, but simply that it depends upon x , or time, or height.

Worked example 5.4

We define a function $g(x) = x^2 + x$.

Find and simplify: (a) $g(2)$ (b) $g(y)$ (c) $g(x+1)$ (d) $g(3x)$ (e) $4g(x-1)-3$

Replace x with 2

$$(a) \quad g(2) = 2^2 + 2 = 6$$

Replace x with y

$$(b) \quad g(y) = y^2 + y$$

Replace x with $x+1$ Don't forget brackets!

$$(c) \quad g(x+1) = (x+1)^2 + (x+1) = x^2 + 2x + 1 + x + 1 = x^2 + 3x + 2$$

Replace x with $3x$ Don't forget brackets!

$$(d) \quad g(3x) = (3x)^2 + (3x) = 9x^2 + 3x$$

Replace x with $x-1$ Don't forget brackets!

$$(e) \quad 4g(x-1) - 3 = 4((x-1)^2 + (x-1)) - 3 \\ = 4(x^2 - 2x + 1 + x - 1) - 3 = 4(x^2 - x) - 3 = 4x^2 - 4x - 3$$

Exercise 5B

1. If $h(x) = 3x^2 - x$, find and simplify:

- | | |
|-------------------------------------|-----------------------|
| (a) (i) $h(3)$ | (ii) $h(7)$ |
| (b) (i) $h(-2)$ | (ii) $h(-1)$ |
| (c) (i) $h(z)$ | (ii) $h(a)$ |
| (d) (i) $h(x+1)$ | (ii) $h(x-2)$ |
| (e) (i) $\frac{1}{2}(h(x) - h(-x))$ | (ii) $3h(x) + 4h(2x)$ |
| (f) (i) $h\left(\frac{1}{x}\right)$ | (ii) $h(\sqrt{x})$ |

Calculations with logarithms were covered in chapter 2.



2. If $g : x \mapsto 1 + \log_{10} x$, find and simplify:

- | | |
|--------------------------|---|
| (a) (i) $g(100)$ | (ii) $g(1\,000\,000)$ |
| (b) (i) $g(0.1)$ | (ii) $g(1)$ |
| (c) (i) $g(y)$ | (ii) $g(z)$ |
| (d) (i) $g(10x)$ | (ii) $g(100y)$ |
| (e) (i) $3g(x) + g(x^2)$ | (ii) $\frac{1}{2}\left(g(x) + g\left(\frac{1}{x}\right)\right)$ |

3. If $u(x) = 3x + 1$ and $v(x) = -\sqrt{x}$, find:

- (a) (i) $u(2) + v(9)$ (ii) $u(1)v(4)$
 (b) (i) $u(x) + v(y)$ (ii) $u(2x+1) - v(4x)$
 (c) (i) $2u(4x) + 3v(4x)$ (ii) $v(x^2 + 1) + xu(2x)$

5C Domain and range

A function tells you what to do with the input, but, to be completely defined, you also need to know what type of input is permitted to go into the function.

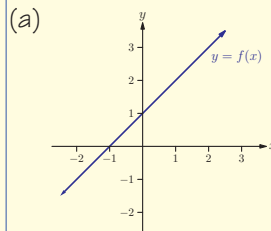
KEY POINT 5.3

The set of allowed input values is called the **domain** of the function. Conventionally we write it after the rule using set notation, interval notation or inequalities.

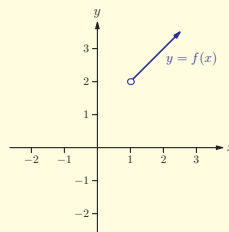
Worked example 5.5

Sketch the graph of $f(x) = x + 1$ over the domain (a) $x \in \mathbb{R}, x > 1$ (b) $x \in \mathbb{Z}$

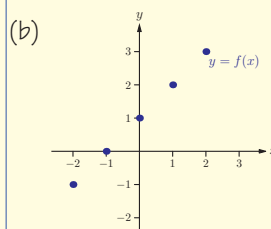
Sketch the graph over the domain $x \in \mathbb{R}$



Discard the part of the graph which is outside of the required domain. Since the endpoint is not included we must label it with an open circle



Use the same original graph as in (a), but this time it only exists at whole numbers, so we label them with closed circles



EXAM HINT

When you are instructed to sketch the graph of $f(x)$, this just means the graph of $y = f(x)$.

It is quite tempting to say that dividing by zero results in infinity. However, doing this leads to some unfortunate consequences, such as all numbers being 'equal'!



If no domain is explicitly mentioned you can assume that the domain is all real numbers.

You may wonder why we would ever need any other domain. There are normally two reasons. We may be modelling a physical situation where the variables can only take particular values; for example if our variable was age of humans we would not expect it to be negative or much beyond 120.

Another possibility is that the mathematical operation we are using cannot handle certain types of numbers. For example if we were looking for the largest prime factor of a number, we would normally only be looking at positive integers.

When working with real numbers, the three most important reasons to restrict the domain are

- you cannot divide by zero
- you cannot square root a negative number
- you cannot take the logarithm of a negative number or zero.

Worked example 5.6

What is the largest possible domain of $h : x \rightarrow \frac{1}{x-2} + \sqrt{x+3}$?

Look for division by zero

Look for square rooting of a negative number

Decide what can therefore be allowed into the function

There will be division by zero when $x - 2 = 0$

There will be square rooting of a negative number when $x + 3 < 0$

$x \geq -3$ and $x \neq 2$

Remember we can also use interval notation to write domains, so in the last example we could write $x \in [-3, 2[\cup]2, \infty[$.

Once we have set the domain, it is interesting to see what values can come out of the function.

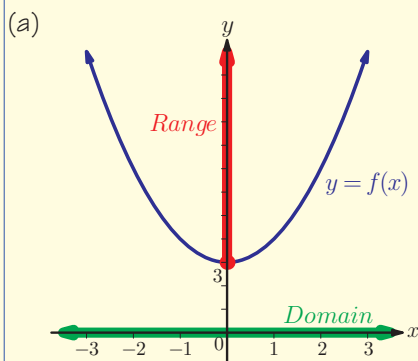
KEY POINT 5.4

The set of all possible outputs of a function is called the **range**. The easiest way of finding this is to sketch the graph (possibly using your GDC). Be aware that the range will depend upon the domain.

Worked example 5.7

Find the range of $f(x) = x^2 + 3$ if the domain is: (a) $x \in \mathbb{R}$ (b) $x > 2$

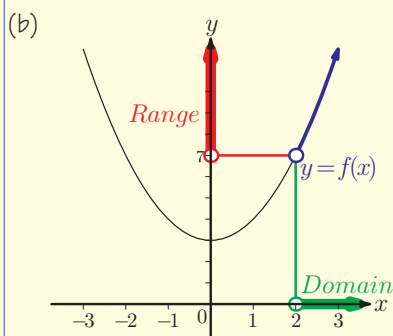
Sketch the graph $y = f(x)$



$$y \geq 3$$

Use the graph to state what y values can occur

Sketch the graph $y = f(x)$



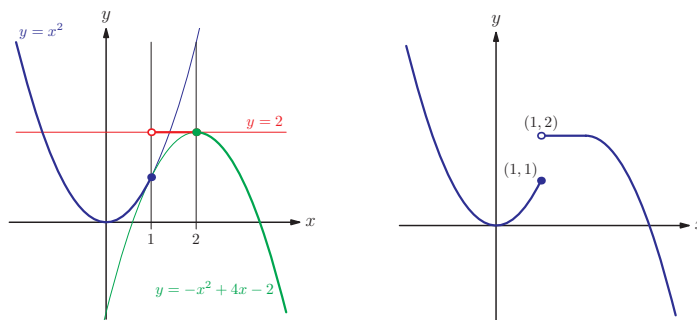
$$y > 7$$

Use the graph to state what y values can occur given domain $x > 2$

A function does not have the same rule over all of its domain. For example, we could define a function $f(x)$ over all of $\mathbb{R}$ as follows:

$$f(x) = \begin{cases} x^2, & x \leq 1 \\ 2, & 1 < x < 2 \\ -x^2 + 4x - 2, & x \geq 2 \end{cases}$$

A function defined in this way is called a **piecewise** function. To sketch the graph of $f(x)$ we need to sketch all three graphs and then select the relevant graph for each part of the domain. Remember to use a closed circle when the end point is included and an open circle when it is not.



As we move from one part of the domain to another, the pieces of the graph may join up (as at $x = 1$ above) or there may be a jump in the y -coordinate (as at $x = 1$). In the former case we say that the function is **continuous** at that point. So the function $f(x)$ above is continuous everywhere except at $x = 1$.

The easiest way to see whether a function is continuous is to sketch its graph. If this is not possible, we can look at the values of the function near the end points of different parts of the domain. For example, $f(1) = 1^2 = 1$, but when x is just larger than 1, $f(x) = 2$. Therefore, $f(x)$ is not continuous at $x = 1$. On the other hand, $f(2) = -2^2 + 4 \times 2 - 2 = 2$, and when x is just smaller than 2, $f(x) = 2$ also. So $f(x)$ is continuous at $x = 2$.

Worked example 5.8

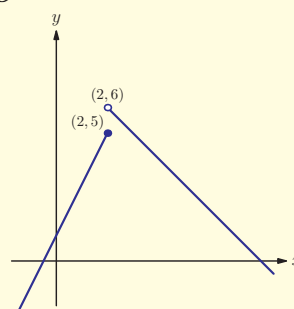
A function is defined by

$$f(x) = \begin{cases} 2x + 1, & x \leq 2 \\ k - x, & x > 2 \end{cases}$$

- If $k = 8$, find the range of $f(x)$.
- Find the value of k for which $f(x)$ is continuous.

To see the range, sketch the graph.
Mark the values at the end points of the different parts of the domain

- When $x = 2$:
 $2x + 1 = 5$
 $8 - x = 6$



continued...

There are no y values between 5 and 6.
5 is included in the range, but 6 is not

For $f(x)$ to be continuous, the two expressions have to have equal values at $x = 2$

The range is $]-\infty, 5] \cup]6, \infty[$

(b) If $f(x)$ is continuous:

$$2x + 1 = k - x \text{ when } x = 2$$

$$\Rightarrow 5 = k - 2$$

$$\Rightarrow k = 7$$

Exercise 5C

1. State the largest real domain of the following functions, and their ranges, over those domains:

(a) $f(x) = 2^x$

(b) $f(x) = a^x, a > 0$

(c) $f(x) = \log_{10} x$

(d) $f(x) = \log_b x, b > 0$

2. Find the domain of the following functions.

(a) (i) $f(x) = \frac{1}{x+2}$ (ii) $f(x) = \frac{5}{x-7}$

(b) (i) $f(x) = \frac{3}{(x-2)(x+4)}$ (ii) $g(x) = \frac{x}{x^2-9}$

(c) (i) $r(y) = \sqrt{y^3-1}$ (ii) $h(x) = \sqrt{x+3}$

(d) (i) $f(a) = \frac{1}{\sqrt{a-1}}$ (ii) $f(x) = \frac{5x}{\sqrt{2-5x}}$

(e) (i) $a(x) = \frac{1}{x} + \frac{2}{x+1}$ (ii) $f(x) = \sqrt{x+1} + \frac{1}{x+2}$

(f) (i) $f(x) = \sqrt{x^2-5}$ (ii) $f(x) = 4\sqrt{x^2+2x-3}$

(g) (i) $f(x) = \sqrt{x} + \frac{1}{x+7} - x^3 + 5$

(ii) $f(x) = e^x + \sqrt{2x+3} - \frac{1}{x^2+4} - 2$

3. Find the range of the following functions.

(a) (i) $f(x) = 7 - x^2, x \in \mathbb{R}$ (ii) $f(x) = x^2 + 3, x \in \mathbb{R}$

(b) (i) $g(x) = x^2 + 3, x \geq 3$ (ii) $h(x) = x + 1, x > 3, x \in \mathbb{Z}$

(c) (i) $f(x) = |x-1|$ (ii) $f(x) = |2x+3|$

(d) (i) $d(x) = \frac{1}{x}, x \geq -1, x \neq 0$ (ii) $q(x) = 3\sqrt{x}, x > 0$

Exponential

functions were covered in chapter 2.

EXAM HINT

If a question asks for 'the domain' of a function, you should answer with the largest possible real domain.

Some of these questions require

solution of quadratic inequalities, covered in Prior learning Section Z on the CD-ROM.



What is the largest possible range and domain of the

function $f(x) = (-2)^x$? This function illustrates why it is important to be careful in deciding how to define a continuous function; an important concept in higher mathematics.

4. For each of the following piecewise functions sketch the graph, find the range and state whether the function is continuous.

(a) (i) $f(x) = \begin{cases} 2+x, & x \leq 0 \\ 3+x, & x > 0 \end{cases}$ (ii) $f(x) = \begin{cases} 2-x, & x < 0 \\ 1-2x, & x \geq 0 \end{cases}$

(b) (i) $f(x) = \begin{cases} x+3, & x < 0 \\ 2e^{-x}, & x \geq 0 \end{cases}$ (ii) $f(x) = \begin{cases} 3e^x, & x \leq 0 \\ x+3, & x > 0 \end{cases}$

(c) (i) $f(x) = \begin{cases} 1, & x \leq -1 \\ x^2, & -1 < x \leq 2 \\ 4, & x > 2 \end{cases}$ (ii) $f(x) = \begin{cases} 0, & x < -2 \\ 4-x^2, & -2 < x \leq 2 \\ 2, & x > 2 \end{cases}$

5. The function f is given by $f(x) = \sqrt{\ln(x-4)}$. Find the domain of the function. [4 marks]

6. Find the largest possible domain of the function

$$f(x) = \frac{4^{\sqrt{x-1}}}{x+2} - \frac{1}{x^2-5x+6} + x^2 + 1. \quad [5 \text{ marks}]$$

7. The function f is defined for all $x \in \mathbb{R}$ by:

$$f(x) = \begin{cases} 3x^2 - 1, & x \leq 2 \\ a - x^2, & x > 2 \end{cases}$$

Find the value of a for which f is continuous. [3 marks]

8. Find the largest possible domain of the function

$$g(x) = \ln(x^2 + 3x + 2). \quad [5 \text{ marks}]$$

9. Find the largest set of values of x such that the function f

given by $f(x) = \sqrt{\frac{8x-4}{x-12}}$ takes real values. [5 marks]

10. (a) State the domain of the function

$$f(x) = \sqrt{x-a} + \ln(b-x) \text{ if}$$

(i) $a < b$

(ii) $a > b$

- (b) Evaluate $f(a)$.

[6 marks]



See Prior learning
Section G on the

CD-ROM for
the definition of
modulus function, $|x|$.



5D Composite functions

We can apply one function to a number and then apply another function to that result.

The resulting rule is called a **composite function**.

KEY POINT 5.5

If we first apply the function g to x and then the function f to the result, we write:

$$f(g(x)) \quad \text{or} \quad fg(x) \quad \text{or} \quad f \circ g(x)$$

It can be useful to refer to $g(x)$ as the **inner function** and $f(x)$ as the **outer function**.

For the composite function $fg(x)$ to exist, the range of $g(x)$ must lie entirely within the domain of $f(x)$, otherwise we would be trying to put values into $f(x)$ which cannot be calculated.

EXAM HINT

Any of these notations are all equally correct, so you can use whichever one suits you. In the exam you must be able to interpret any of them. Remember the correct order: The function nearest to x acts first!

Worked example 5.9

If $f(x) = x^2$ and $g(x) = x - 3$, find

(a) $f \circ g(1)$ (b) $fg(x)$ (c) $gf(x)$

We need to evaluate $g(1)$ and then apply f to the result

$$\begin{aligned} \text{(a)} \quad g(1) &= 1 - 3 = -2 \\ f(-2) &= (-2)^2 \\ &= 4 \\ \therefore f(g(1)) &= 4 \end{aligned}$$

Replace x in $f(x)$ with the expression for $g(x)$

$$\text{(b)} \quad f(g(x)) = f(x - 3) = (x - 3)^2 = x^2 - 6x + 9$$

Replace x in $g(x)$ with the expression for $f(x)$

$$\text{(c)} \quad g(f(x)) = g(x^2) = x^2 - 3$$

EXAM HINT

Notice that to answer part (a) we did not need to work out the general expression for $f \circ g(x)$.

Notice that $f(g(x))$ and $g(f(x))$ are not the same function.

It is more difficult to recover one of the original functions from a composite function. The best way to do this is to use a substitution.

Worked example 5.10

If $f(x+1) = 4x^2 + x$, find $f(x)$.

Substitute $y = \text{the inner function}$

Rearrange to get x as the subject

Replace all instances of x

We were asked to write the answer in terms of x

$$y = x + 1$$

$$x = y - 1$$

$$\begin{aligned} f(y) &= 4(y-1)^2 + (y-1) \\ &= 4y^2 - 8y + 4 + y - 1 \\ &= 4y^2 - 7y + 3 \end{aligned}$$

$$f(x) = 4x^2 - 7x + 3$$

Exercise 5D

- If $f(x) = x^2 + 1$ and $g(x) = 3x + 2$, find:
 - $g(f(0))$
 - $fg(1)$
 - $gg(x)$
 - $f \circ g(x)$
 - $gg(\sqrt{a} + 1)$
 - $f \circ f(y - 1)$
 - $gfg(y)$
 - $gfg(z)$
- Find $f(x)$ given the following conditions:
 - $f(2a) = 4a^2$
 - $f\left(\frac{b}{3}\right) = \frac{b^3}{27}$
 - $f(x+1) = 3x - 2$
 - $f(x-2) = x^2 + x$
 - $f(1-y) = 5 - y$
 - $f(y^3) = y^2$
 - $f(e^k) = \ln k$
 - $f(3n+2) = \ln(n+1)$
- If $f(x) = x^2 + 1$ and $g(x) = 3x + 2$, solve $fg(x) = gf(x)$. [4 marks]

4. If $f(x) = 3x + 1$ and $g(x) = \frac{x}{x^2 + 25}$, solve $gf(x) = 0$.
[5 marks]



5. Functions g and h are defined by

$$g(x) = \sqrt{x} \text{ and } h(x) = \frac{2x-3}{x+1} \quad (x \neq -1).$$

- (a) Find the range of h .
(b) Solve the equation $h(x) = 0$.
(c) Find the domain and range of $g \circ h$. [6 marks]

6. The function f is defined by $f: x \mapsto x^3$. Find an expression for $g(x)$ in terms of x in each of the following cases.

- (a) $f \circ g(x) = 2x + 3$
(b) $g \circ f(x) = 2x + 3$ [6 marks]

7. Functions f and g are defined by $f(x) = \sqrt{x^2 - 2x}$ and $g(x) = 3x + 4$. The composite function $f \circ g$ is undefined for $x \in]a, b[$.

- (a) Find the value of a and the value of b .
(b) Find the range of $f \circ g$. [7 marks]

8. If $f(x) = x - 1$, $x > 3$ and $g(x) = x^2$, $x \in \mathbb{R}$:

- (a) Show that $g \circ f$ exists and $f \circ g$ does not.
(b) Find the largest possible domain for g so that $f \circ g$ is defined. [6 marks]

9. Let f and g be two functions. Given that $f \circ g(x) = \frac{x+2}{3}$ and $g(x) = 2x + 5$, find $f(x - 1)$. [6 marks]

5E Inverse functions

Functions transform an input into an output, but sometimes we need to reverse this process to be able to say which input produced a particular output. When this is possible, it is done by finding the **inverse function**, usually labelled f^{-1} .

For example, if $f(x) = 3x$ then $f^{-1}(12)$ is a number which when put into f produces output 12. In other words, we are looking for a number x such that $f(x) = 12$, hence $f^{-1}(12) = 4$.

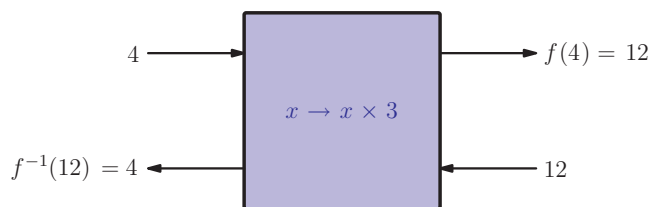
EXAM HINT

Make sure you do not get confused about this notation.

With numbers, the superscript $'-1'$ denotes reciprocal,

e.g. $x^{-1} = \frac{1}{x}$, $3^{-1} = \frac{1}{3}$

With functions, f^{-1} denotes the inverse function of f .



To find the inverse function you rearrange the formula to find the input (x) in terms of the output (y).

KEY POINT 5.6

To find the inverse function $f^{-1}(x)$ from an expression for $f(x)$

1. start off with $y = f(x)$
2. rearrange to get x (the input) in terms of y (the output) i.e. make x the subject
3. but this is $f^{-1}(y)$ and we can restate with x instead of y to show $f^{-1}(x)$.

Worked example 5.11

Find the inverse function of $f(x) = \frac{1+x}{3-x}$.

Set $y = f(x)$

$$y = \frac{1+x}{3-x}$$

Make x the subject

$$\begin{aligned} y(3-x) &= 1+x \\ 3y - yx &= 1+x \\ 3y - 1 &= x + xy \\ 3y - 1 &= x(1+y) \end{aligned}$$

Restate with x instead of y

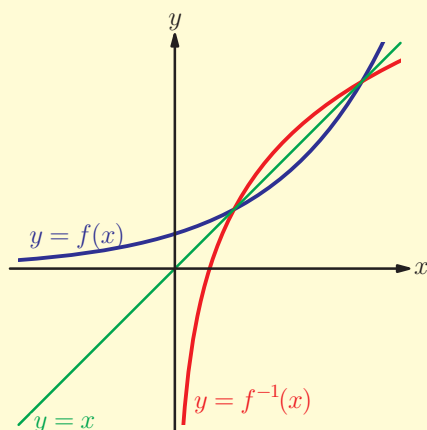
$$\begin{aligned} x &= \frac{3y-1}{1+y} \\ \therefore f^{-1}(y) &= \frac{3y-1}{1+y} \\ f^{-1}(x) &= \frac{3x-1}{1+x} \end{aligned}$$

Once we know how to find inverse functions, there are some important facts we need to know about them:

- When we are finding the inverse function we switch the inputs and the outputs, so on the graph we switch the x - and y -axis.

KEY POINT 5.7

The graph of $y = f^{-1}(x)$ is a reflection of the graph of $y = f(x)$ in the line $y = x$.



EXAM HINT

See Calculator sheet 7 on the CD-ROM to see how you can sketch the graph of the inverse function on your calculator.



- When you do and then undo a function you get back to where you started.

KEY POINT 5.8

$$f(f^{-1}(x)) = f^{-1}(f(x)) = x$$

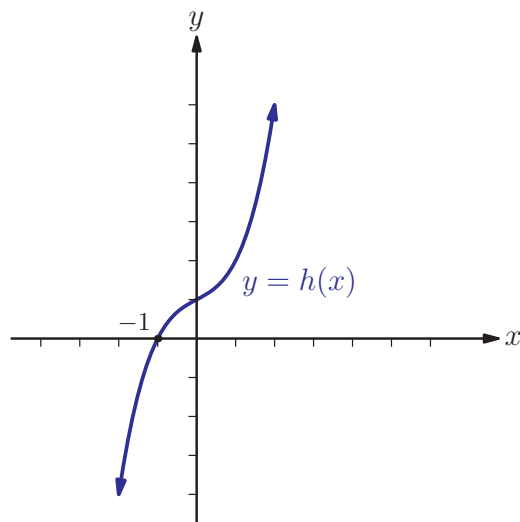
There are two pieces of vocabulary which you need to know:

A function which does not change the input is called an **identity function**, so the key point above says that the composition of a function and its inverse is the identity function.

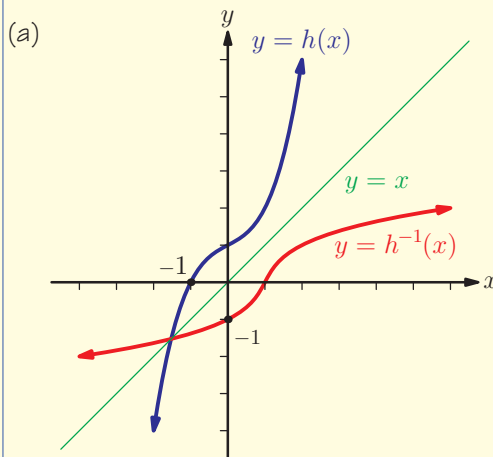
For some functions, such as $f(x) = \frac{3}{x}$ or $g(x) = 5 - x$ the inverse function is the same as the original function so that $f \circ f(x)$ is the identity function. In this case the function is called a **self-inverse function**.

Worked example 5.12

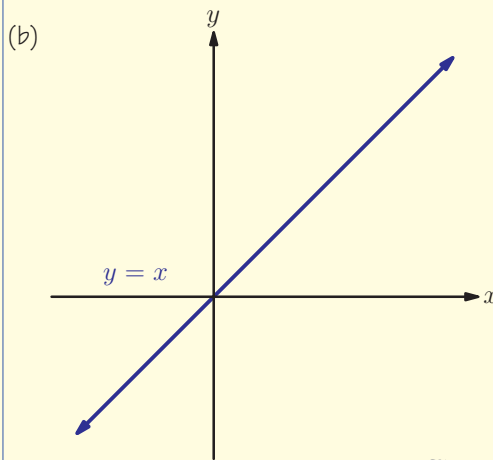
The graph of $y = h(x)$ is shown below. Sketch the graphs of $y = h^{-1}(x)$ and $y = h \circ h^{-1}(x)$.



The graph of $y = h^{-1}(x)$ is a reflection in the line $y = x$ of $y = h(x)$



Simplify $y = h \circ h^{-1}(x)$ to $y = x$ and then sketch this over the domain of h



- The reflection in the line $y = x$ swaps the domain and the range of a function (because it swaps x - and y -coordinates).

KEY POINT 5.9

The domain of $f^{-1}(x)$ is the same as the range of $f(x)$.

The range of $f^{-1}(x)$ is the same as the domain of $f(x)$.

- All functions have inverse relations, but these inverses are not necessarily themselves functions. Since an inverse function is a reflection in the line $y = x$, for the result to pass the vertical line test the original function must pass the horizontal line test. But as we saw in Section 5A, this means it must be a one-to-one function.

KEY POINT 5.10

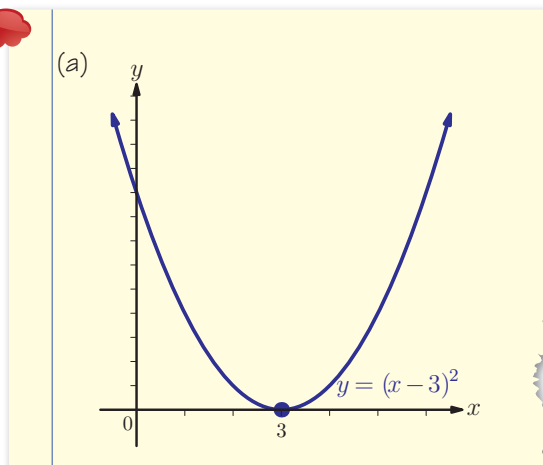
Only one-to-one functions have inverse functions.

This leads us to one of the most important uses of domains. By restricting the domain we can turn any function into a one-to-one function, which allows us to find its inverse function.

Worked example 5.13

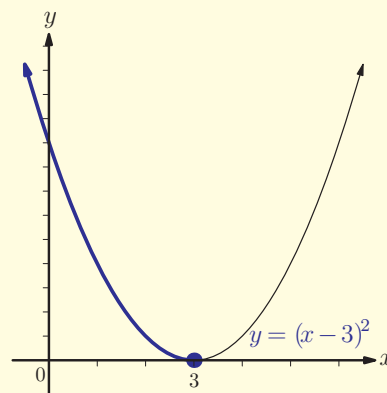
- Find the largest value of k such that the function $f(x) = (x - 3)^2$, $x \leq k$ is one-to-one.
- Find $f^{-1}(x)$ for this value of k and state its range.

Sketch the graph of $y = (x - 3)^2$, $x \in \mathbb{R}$



continued...

Eliminate the points towards the right of the graph which cause the horizontal line test to fail



Decide which section remains

$$x \leq 3$$

Follow the standard procedure for finding inverse functions

$$\begin{aligned} (b) \quad y &= (x-3)^2 \\ \pm\sqrt{y} &= x-3 \\ x &= 3 \pm \sqrt{y} \end{aligned}$$

Use the fact that $x \leq 3$ to decide which root to take

$$\begin{aligned} \text{Since } x &\leq 3 \\ x &= 3 - \sqrt{y} \end{aligned}$$

Write $f^{-1}(x) = \dots$

$$f^{-1}(x) = 3 - \sqrt{x}$$

The range of f^{-1} is the domain of f

The range of f^{-1} is $y \leq 3$

EXAM HINT

You can check that your expression for $f^{-1}(x)$ is correct by sketching f and f^{-1} on the same graph and looking for symmetry. Make sure you use the correct domain.

Exercise 5E

1. Find $f^{-1}(x)$ if:

(a) (i) $f(x) = 3x + 1$

(ii) $f(x) = 7x - 3$

(b) (i) $f(x) = \frac{2x}{3x-2}, x \neq \frac{2}{3}$

(ii) $f(x) = \frac{x}{2x+1}, x \neq -\frac{1}{2}$

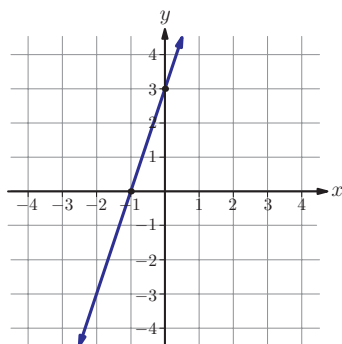
(c) (i) $f(x) = \frac{x-a}{x-b}, x \neq b$

(ii) $f(x) = \frac{ax-1}{bx-1}, x \neq \frac{1}{b}$

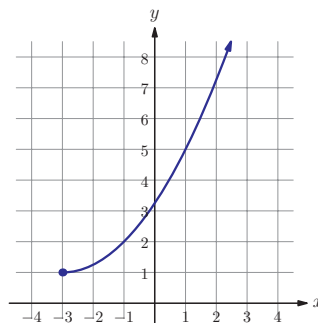
- (d) (i) $f(a) = 1 - a$ (ii) $f(y) = 3y + 2$
- (e) (i) $f(x) = \sqrt{3x - 2}, x \geq \frac{2}{3}$ (ii) $f(x) = \sqrt{2 - 5x}, x \leq \frac{2}{5}$
- (f) (i) $f(x) = \ln(1 - 5x), x < 0.2$ (ii) $f(x) = \ln(2x + 2), x > -1$
- (g) (i) $f(x) = 7e^{\frac{x}{2}}$ (ii) $f(x) = 9e^{10x}$
- (h) (i) $f(x) = x^2 - 10x + 6, x < 5$ (ii) $f(x) = x^2 + 6x - 1, x > 0$

2. Sketch the inverse relations of the following functions.

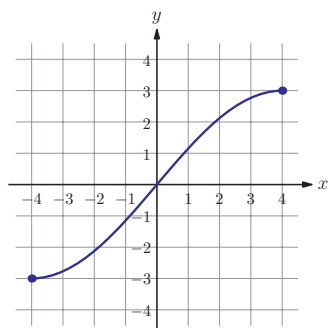
(a)



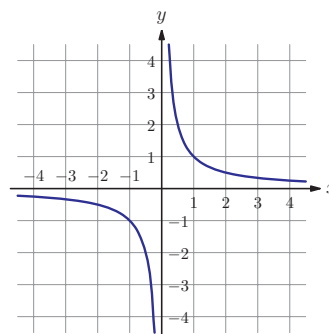
(b)



(c)



(d)



3. For each of the following functions, find and simplify $f \circ f(x)$ and hence state if the function is self-inverse.

- (a) (i) $f: x \mapsto -x$ (ii) $f: x \mapsto x$
- (b) (i) $f: x \mapsto 2x$ (ii) $f: x \mapsto \frac{x}{-4}$
- (c) (i) $f: x \mapsto 10 + x$ (ii) $f: x \mapsto 1 + x$
- (d) (i) $f: x \mapsto 7 - x$ (ii) $f: x \mapsto 1 - x$
- (e) (i) $f: x \mapsto \frac{10}{x}$ (ii) $f: x \mapsto \frac{-3}{x}$
- (f) (i) $f: x \mapsto \frac{3}{x^2}$ (ii) $f: x \mapsto \frac{1}{x^3}$
- (g) (i) $f: x \mapsto \sqrt{x}$ (ii) $f: x \mapsto \sqrt[3]{x}$

$$(h) (i) f: x \mapsto \frac{1}{1-x}$$

$$(ii) f: x \mapsto \frac{1}{1+x}$$

$$(i) (i) f: x \mapsto \frac{2x+3}{x-2}$$

$$(ii) f: x \mapsto \frac{1-3x}{3+x}$$

4. Consider the function $f(x) = \sqrt{x^2}$. Over what domain is this an identity function?

5. Below is a table giving selected values of the one-to-one function $f(x)$:

x	-1	0	1	2	3	4
$f(x)$	-4	-1	3	0	7	2

(a) Evaluate $ff(2)$.

(b) Evaluate $f^{-1}(3)$. [4 marks]

6. The function f is defined by $f: x \mapsto \sqrt{3-2x}$, $x \leq \frac{3}{2}$.

Evaluate $f^{-1}(7)$. [4 marks]

7. Given that $f(x) = 3e^{2x}$, find the inverse function $f^{-1}(x)$. [4 marks]

8. Given functions $f: x \mapsto 2x+3$ and $g: x \mapsto x^3$, find the function $(f \circ g)^{-1}$. [5 marks]

9. The functions f and g are defined by: $f: x \mapsto e^{2x}$, $g: x \mapsto x+1$:

(a) Calculate $f^{-1}(3) \times g^{-1}(3)$.

(b) Show that $(f \circ g)^{-1}(3) = \ln\sqrt{3} - 1$. [6 marks]

10. Let $f(x) = \sqrt{x}$, and $g(x) = 2^x$. Solve the equation:

$$f^{-1} \circ g(x) = 0.25. \quad [5 \text{ marks}]$$

11. The function f is defined for $x \leq 0$ by $f(x) = \frac{x^2-4}{x^2+9}$.

Find an expression for $f^{-1}(x)$. [5 marks]

12. For the following functions, find the value of k which gives the largest possible domain such that the inverse function exists. For this domain, find the inverse function.

(a) $y = x^2$, $x \leq k$

(b) $y = (x+1)^2 + 2$, $x \geq k$

(c) $y = |x|$, $x \leq k$ [6 marks]

13. Let $f(x) = \ln(x-1) + \ln 3$, for $x > 1$.

(a) Find $f^{-1}(x)$.

(b) Let $g(x) = e^x$. Find $(g \circ f)(x)$, giving your answer in the form $ax + b$, where $a, b \in \mathbb{Z}$. [7 marks]

14. A piecewise function is defined by:

$$f(x) = \begin{cases} 2 + (x-1)^2, & x < 1 \\ k - (x-1)^2, & x \geq 1 \end{cases}$$

- (a) Find the largest value of k for which f is a one-to-one function.

- (b) When $k = 0$:

(i) Find the range of f .

(ii) Find an expression for $f^{-1}(x)$. [8 marks]

15. A function is called *self-inverse* if $f(x) = f^{-1}(x)$ for all x in the domain.

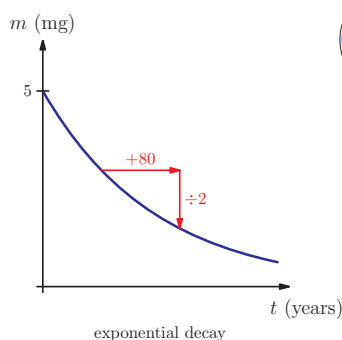
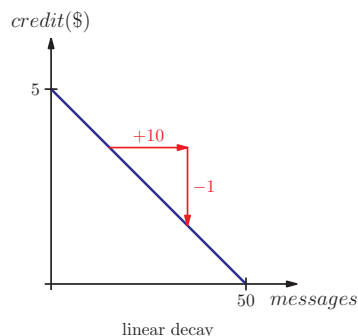
- (a) Show that $f(x) = \frac{1}{x}$, $x \neq 0$ is a self-inverse function.

- (b) Find the value of the constant k so that

$$g(x) = \frac{3x-5}{x+k}, x \neq k \text{ is a self-inverse function. [8 marks]}$$

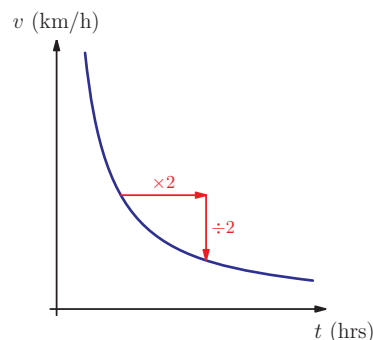
5F Rational functions

There are many situations where one quantity decreases as another increases. For example, the amount of your phone credit decreases as the number of text messages you send increases. As the number of messages increases by a fixed number, the credit decreases by a fixed amount; this is called **linear decay**. Another example is radioactive decay, where the amount of a radioactive substance halves in a fixed time period (called half-life). So as the time increases by a fixed number, the amount of substance decreases by a fixed factor; this is called **exponential decay**.



Exponential functions are covered in chapter 2, and linear functions in Section R of Prior learning.

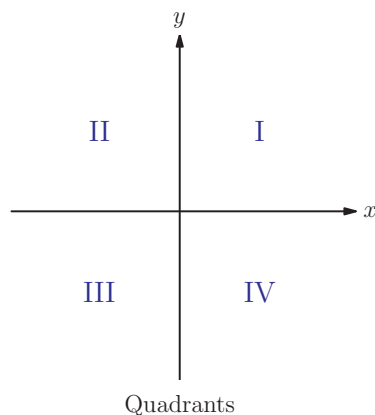
In this chapter we will look at a third type of decay, called **inverse proportion**, where as one quantity increases by a fixed factor, another decreases by the same factor. For example, if you double your speed, the amount of time to travel a given distance will halve. If the total distance travelled is 12 km, then the equation for time (in hours) in terms of speed (in km/h) is $t = 12/v$. This is an example of a **reciprocal function**.



EXAM HINT

The reciprocal of a real number $x \neq 0$ is $\frac{1}{x}$. For example, the reciprocal of -2 is $-\frac{1}{2}$ and the reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$.

Graphs of reciprocal functions all have the same shape, called a **hyperbola**. This curve consists of two parts and has the axis as the asymptotes. This means that the function is not defined for $x = 0$, and that as x gets very large (positive or negative), y approaches zero. Therefore neither x nor y can equal zero. The two parts of the curve can be either in the first and third, or the second and fourth quadrants, depending on the sign of k .

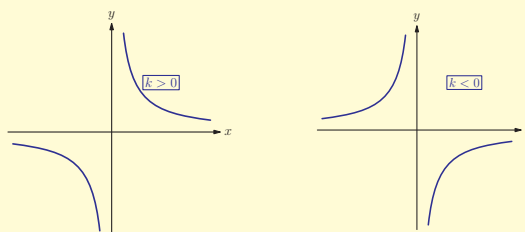


KEY POINT 5.11

A reciprocal function has the form $f(x) = \frac{k}{x}$.

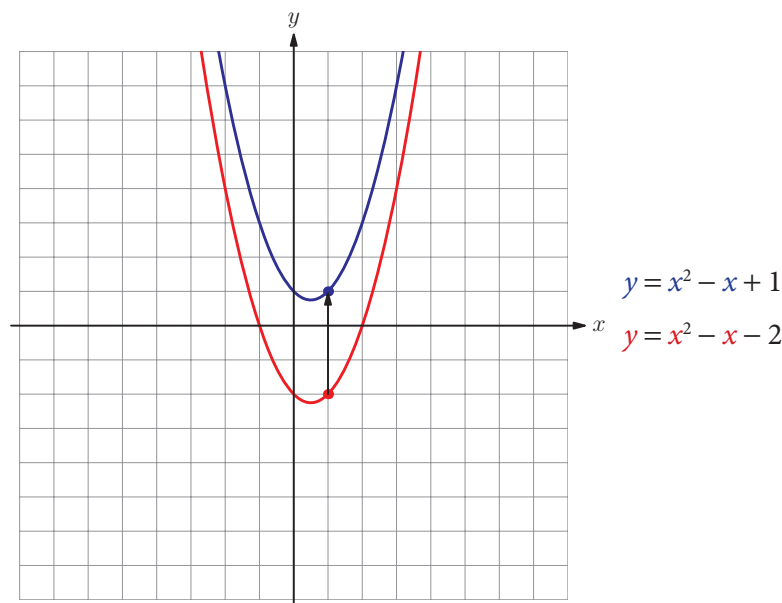
The domain of f is $x \neq 0$ and the range is $y \neq 0$.

The graph of $f(x)$ is a hyperbola.



6A Translations

Compare the graphs of two functions that only differ by a constant:



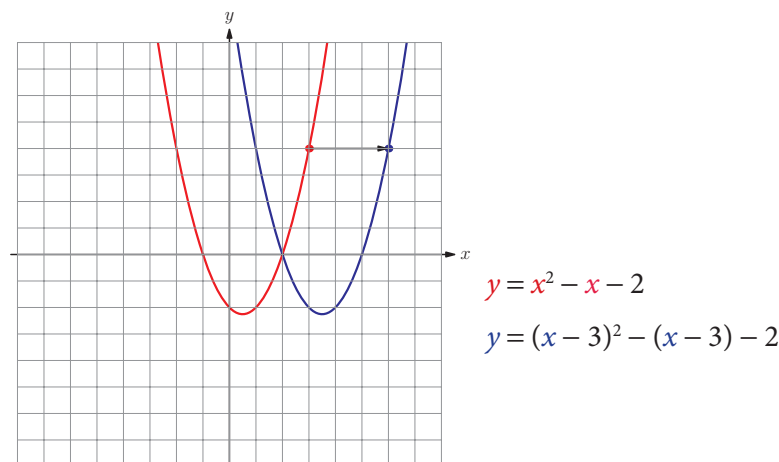
When the x -coordinates on the two graphs are the same (where $x = x$) the y -coordinates differ by 3 ($y = y + 3$). We can interpret this as meaning that at the same x -coordinates the blue graph is three units above the red graph – it has been translated vertically.

KEY POINT 6.1

The graph $y = f(x) + c$ is the graph of $y = f(x)$ translated up by c units.

If c is negative the graph is translated down.

In the next example, the blue function is obtained from the red function by replacing x by $x - 3$.



Mathematics is often thought to be a 'universal' language and it is not easy to find examples where our language influences mathematics. However, the educational researcher Bill Barton suggests that the Cartesian coordinate system we are using to visualise functions is a reflection of our own culture, with everything being measured relative to a fixed origin. He suggests that Maori mathematics, where it is considered rude to describe a position relative to just one person, leads to a different coordinate system which is described on Supplementary sheet 4 'Coordinate systems and graphs'.



A translation can also be described using a

vector: $\begin{pmatrix} a \\ b \end{pmatrix}$ means

translate a units horizontally and b units vertically. You will learn more about vectors in chapter 13.

If $x = x$ there is nothing useful we can say about the relationship between y and y .

However, a way of making $y = y$ is to have $x = x - 3$ or equivalently $x + 3 = x$. We can interpret this as meaning that when the two graphs have the same height, the blue graph is three units to the right of the red graph; it has been translated horizontally.

KEY POINT 6.2

The graph $y = f(x + d)$ is the graph of $y = f(x)$ translated left by d units. If d is negative the graph is translated right.

Worked example 6.1

The graph of $y = x^2 + 2x$ is translated 5 units to the left.

Find the equation of the resulting graph in the form $y = ax^2 + bx + c$.

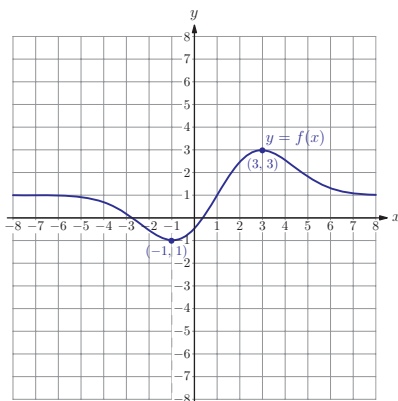
Relate the transformation to function notation

Replace all instances of x in the equation by $(x + 5)$

If $f(x) = x^2 + 2x$ then the new graph is $y = f(x + 5)$

$$\begin{aligned} y &= (x + 5)^2 + 2(x + 5) \\ &= x^2 + 12x + 35 \end{aligned}$$

Exercise 6A



1. The graph shows a function $y = f(x)$. Sketch the graphs of the following functions, including the positions of the minimum and maximum points.

- | | |
|--------------------------|-----------------------|
| (a) (i) $y = f(x) + 3$ | (ii) $y = f(x) + 5$ |
| (b) (i) $y = f(x) - 7$ | (ii) $y = f(x) - 0.5$ |
| (c) (i) $y = f(x + 2)$ | (ii) $y = f(x + 4)$ |
| (d) (i) $y = f(x - 1.5)$ | (ii) $y = f(x - 2)$ |

2. Find the equation of the graph after the given transformation is applied.

- | |
|--|
| (a) (i) $y = 3x^2$ after a translation of 3 units vertically up. |
| (ii) $y = 9x^3$ after a translation of 7 units vertically down. |
| (b) (i) $y = 7x^3 - 3x + 6$ after a translation of 2 units down. |
| (ii) $y = 8x^2 - 7x + 1$ after a translation of 5 units up. |

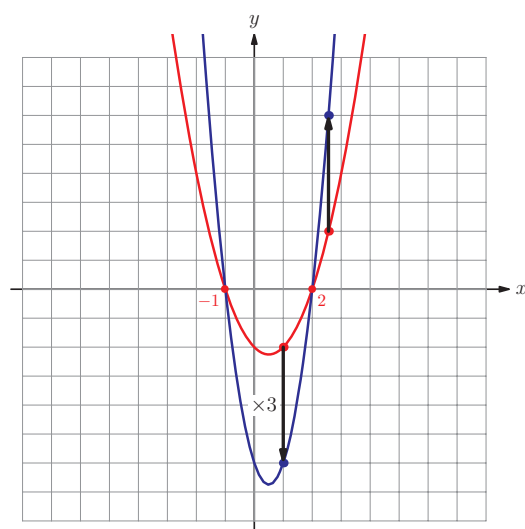
- (c) (i) $y = 4x^2$ after a translation of 5 units to the right.
 (ii) $y = 7x^2$ after a translation of 3 units to the left.
 (d) (i) $y = 3x^3 - 5x^2 + 4$ after a translation of 4 units to the left.
 (ii) $y = x^3 + 6x + 2$ after a translation of 3 units to the right.

3. Find the required translations.

- (a) (i) Transform the graph $y = x^2 + 3x + 7$ to the graph $y = x^2 + 3x + 2$.
 (ii) Transform the graph $y = x^3 - 5x$ to the graph $y = x^3 - 5x - 4$.
 (b) (i) Transform the graph $y = x^2 + 2x + 7$ to the graph $y = (x+1)^2 + 2(x+1) + 7$.
 (ii) Transform the graph $y = x^2 + 5x - 2$ to the graph $y = (x+5)^2 + 5(x+5) - 2$.
 (c) (i) Transform the graph $y = e^x + x^2$ to the graph $y = e^{x-4} + (x-4)^2$.
 (ii) Transform the graph $y = \log(3x) - \sqrt{4x}$ to the graph $y = \log(3(x-5)) - \sqrt{4(x-5)}$.
 (d) (i) Transform the graph $y = \ln(4x)$ to the graph $y = \ln(4x+12)$.
 (ii) Transform the graph $y = \sqrt{2x+1}$ to the graph $y = \sqrt{2x-3}$.

6B Stretches

With the following two graphs, one function is 3 times the other.



$$y = x^2 - x - 2$$

$$y = 3(x^2 - x - 2)$$

When the x -coordinates on the two graphs are the same (where $x = x$), the y -coordinate of the blue graph is three times as large ($y = 3y$). We can interpret this as meaning that, at the same x -coordinates, the blue graph is three times as high as the red graph; it has been stretched vertically.

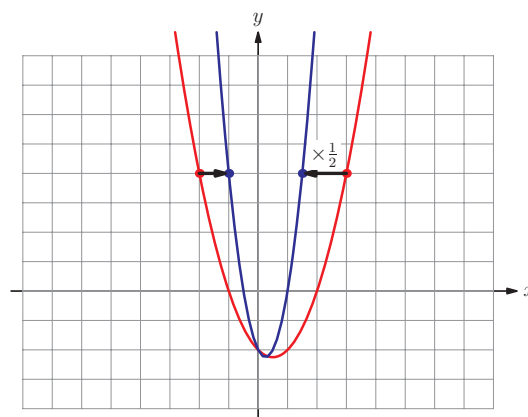
KEY POINT 6.3

The graph $y = pf(x)$, where $p > 0$, is the graph of $y = f(x)$ stretched vertically relative to the x -axis (away from) with scale factor p . If $0 < p < 1$, then $y = f(x)$ is compressed vertically relative to the x -axis (towards). If $p < 0$ the scale factor is negative ($-p$) and it might be easier to think of the transformation as a stretch/compression by scale factor p followed by reflection in the x -axis.

In the next example, the second function is obtained by replacing x by $2x$.

$$y = x^2 - x - 2$$

$$y = (2x)^2 - (2x) - 2$$



If $x = x$ there is nothing useful we can say about the relationship between y and y . However, a way of making $y = y$ is to have $x = 2x$ or equivalently $\frac{x}{2} = x$. We can interpret this as meaning that when the two graphs have the same height, the distance from the y -axis of the blue graph is half the distance of the red graph; it has been stretched horizontally.

KEY POINT 6.4

The graph $y = f(qx)$ is the graph of $y = f(x)$ stretched horizontally relative to the y -axis by scale factor $\frac{1}{q}$. This can be considered a compression relative to the y -axis (towards) when $q > 0$. When $0 < q < 1$, it is considered a stretch relative to the y -axis (away from) and when $q < 0$, the scale factor is negative $\left(-\frac{1}{q}\right)$ and it is easier to think of the transformation as a stretch/compression by scale factor $\frac{1}{q}$ followed by a reflection in the y -axis.

Worked example 6.2

Describe a transformation which transforms the graph of $y = \ln x - 1$ to the graph of $y = \ln x^4 - 4$.

Try to relate the two equations using function notation.

None of the transformations we know involve raising x to a power, so think of a different way to write $\ln x^4$

Relate the function notation to the transformation

$$\begin{aligned}\text{Let } f(x) &= \ln x - 1 \\ \ln x^4 - 4 &= 4 \ln x - 4 \\ &= 4f(x)\end{aligned}$$

It is a vertical stretch with scale factor 4

Exercise 6B

1. Alongside is the graph of $y = f(x)$.

Sketch the graphs of the following functions, including the positions of the minimum and maximum points.

(a) (i) $y = 3f(x)$ (ii) $y = 5f(x)$

(b) (i) $y = \frac{f(x)}{4}$ (ii) $y = \frac{f(x)}{2}$

(c) (i) $y = f(2x)$ (ii) $y = f(6x)$

(d) (i) $y = f\left(\frac{2x}{3}\right)$ (ii) $y = f\left(\frac{5x}{6}\right)$

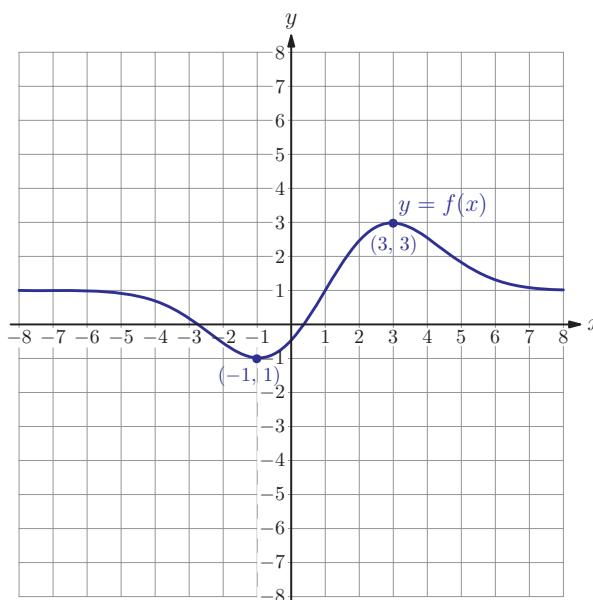
2. Find the equation of the graph after the given transformation is applied:

(a) (i) $y = 3x^2$ after a vertical stretch factor 7 relative to the x -axis

(ii) $y = 9x^3$ after a vertical stretch factor 2 relative to the x -axis

(b) (i) $y = 7x^3 - 3x + 6$ after a vertical stretch factor $\frac{1}{3}$ relative to the x -axis

(ii) $y = 8x^2 - 7x + 1$ after a vertical stretch factor $\frac{4}{5}$ relative to the x -axis



- (c) (i) $y = 4x^2$ after a horizontal stretch factor 2 relative to the y -axis
- (ii) $y = 7x^2$ after a horizontal stretch factor 5 relative to the y -axis
- (d) (i) $y = 3x^3 - 5x^2 + 4$ after a horizontal stretch factor $\frac{1}{2}$ relative to the y -axis
- (ii) $y = x^3 + 6x + 2$ after a horizontal stretch factor $\frac{2}{3}$ relative to the y -axis

3. Describe the following transformations:

- (a) (i) Transforming the graph $y = x^2 + 3x + 7$ to the graph $y = 4x^2 + 12x + 28$
- (ii) Transforming the graph $y = x^3 - 5x$ to the graph $y = 6x^3 - 30x$
- (b) (i) Transforming the graph $y = x^2 + 2x + 7$ to the graph $y = (3x)^2 + 2(3x) + 7$
- (ii) Transforming the graph $y = x^2 + 5x - 2$ to the graph $y = (4x)^2 + 5(4x) - 2$
- (c) (i) Transforming the graph $y = e^x + x^2$ to the graph $y = e^{\frac{x}{2}} + \left(\frac{x}{2}\right)^2$
- (ii) Transforming the graph $y = \log(3x) - \sqrt{4x}$ to the graph $y = \log\left(\frac{3x}{5}\right) - \sqrt{\frac{4x}{5}}$
- (d) (i) Transforming the graph $y = \ln(4x)$ to the graph $y = \ln(12x)$
- (ii) Transforming the graph $y = \sqrt{2x+1}$ to the graph $y = \sqrt{x+1}$

6C Reflections

Compare these pairs of graphs and their equations.

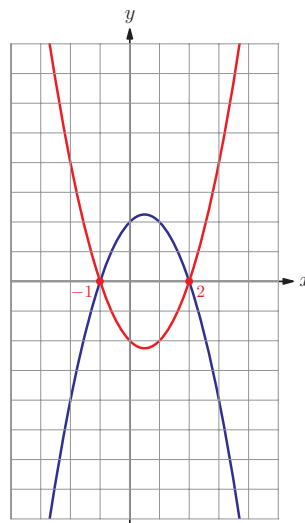
When the x -coordinates on the two graphs are the same (where $x = x$) the y -coordinates are the negative equivalent of each other ($y = -y$). We can interpret this as meaning that, at the same x -coordinates, the blue graph is the same vertical distance relative to the x -axis but in the opposite direction to the red graph; it has been reflected vertically.

KEY POINT 6.5

The graph $y = -f(x)$ is the graph of $y = f(x)$ reflected in the x -axis.

$$y = x^2 - x - 2$$

$$y = -(x^2 - x - 2)$$



In the second example, x has been replaced by $-x$.

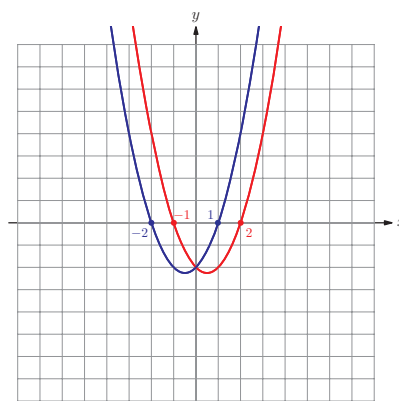
If $x = x$ there is nothing useful we can say about the relationship between y and y .

However, a way of making $y = y$ is to have $x = -x$.

We can interpret this as meaning that when the heights of the two graphs are the same, the blue graph is on the opposite side of the y -axis to the red graph – it has been reflected horizontally.

KEY POINT 6.6

The graph $y = f(-x)$ is the graph of $y = f(x)$ reflected in the y -axis.



$$y = x^2 - x - 2$$

$$y = (-x)^2 - (-x) - 2$$

Worked example 6.3

The graph of $y = f(x)$ has a single maximum point with coordinates $(4, -3)$. Find the coordinates of the maximum point on the graph of $y = f(-x)$.

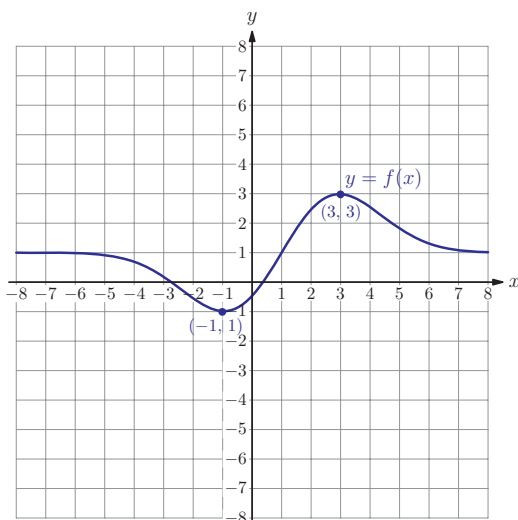
Relate the function notation to a transformation

Reflection in the y -axis leaves y -coordinates unchanged and changes x to $-x$

The transformation taking $y = f(x)$ to $y = f(-x)$ is a reflection in the y -axis

The maximum point is $(-4, -3)$

Exercise 6C



- Alongside is the graph of $y = f(x)$. Sketch the graph of the following functions, including the position of the minimum and maximum points.
 - $y = -f(x)$
 - $y = f(-x)$
- Find the equation of the graph after the given transformation is applied.
 - $y = 3x^2$ after reflection in the x -axis
 - $y = 9x^3$ after reflection in the x -axis
 - $y = 7x^3 - 3x + 6$ after reflection in the x -axis
 - $y = 8x^2 - 7x + 1$ after reflection in the x -axis
 - $y = 4x^2$ after reflection in the y -axis
 - $y = 7x^3$ after reflection in the y -axis
 - $y = 3x^3 - 5x^2 + 4$ after reflection in the y -axis
 - $y = x^3 + 6x + 2$ after reflection in the y -axis
- Describe the following transformations:
 - Transform the graph $y = x^2 + 3x + 7$ to the graph $y = -x^2 - 3x - 7$
 - Transform the graph $y = x^3 - 5x$ to the graph $y = 5x - x^3$
 - Transform the graph $y = x^2 + 2x + 7$ to the graph $y = x^2 - 2x + 7$
 - Transform the graph $y = x^2 - 5x - 2$ to the graph $y = x^2 + 5x - 2$
 - Transform the graph $y = e^x + x^2$ to the graph $y = e^{-x} + x^2$
 - Transform the graph $y = \log(3x) - \sqrt{4x}$ to the graph $y = \sqrt{4x} - \log(3x)$
 - Transform the graph $y = \ln(4x)$ to the graph $y = \ln(-4x)$
 - Transform the graph $y = \sqrt{2x+1}$ to the graph $y = \sqrt{1-2x}$

6D Modulus transformations

The modulus (absolute value) leaves positive numbers unchanged but reverses the sign of negative numbers. In this section we consider the effect of applying modulus transformations to graphs.



You can think of the modulus function as giving the distance of a number away from the zero on the number line. This allows the modulus function to be applied to many other geometric situations. But the distance between two objects is not always easy to define – what length gives the ‘distance’ between two points on the surface of the Earth? This introduces the idea of a *metric* – a way of defining distances between objects. You will not be surprised by the *Euclidean Metric* for distances in flat space, but there are some amazing non-Euclidean metrics including the *Minkowski Metric* for finding the distance between two points in space-time in the theory of relativity.

The function $|f(x)|$ may be described as:

$$|f(x)| = \begin{cases} f(x) & \text{when } f(x) \geq 0 \\ -f(x) & \text{when } f(x) < 0 \end{cases}$$

Hence the graph of $y = |f(x)|$ will be identical to that of $y = f(x)$ wherever $f(x)$ is positive, and will be the same as $y = -f(x)$ whenever $f(x)$ is negative.

KEY POINT 6.7

To draw $y = |f(x)|$, draw the graphs of both $y = f(x)$ and $y = -f(x)$, and take only those parts which are above the x -axis.



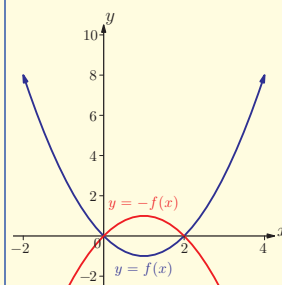
See Prior learning G on the CD-ROM for an introduction to the modulus.

The idea of modulus will be extended to vectors in chapter 13 and complex numbers in chapter 15.

Worked example 6.4

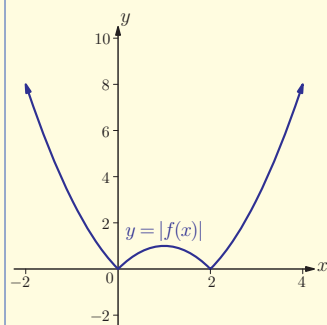
If $f(x) = x(x - 2)$, draw the graph of $y = |f(x)|$.

Lightly draw both $y = f(x)$ and $y = -f(x)$



continued . . .

Restrict the values of y to non-negative



The function $f(|x|)$ will have the same output value for $-x$ as for x because $|x| = |-x|$. Thus:

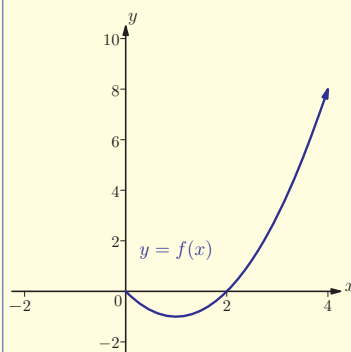
KEY POINT 6.8

The graph of $y = f(|x|)$ can be drawn directly from the graph of $y = f(x)$ by plotting $y = f(x)$ for $x \geq 0$, and then applying the mirror-image of that plot for negative values of x .

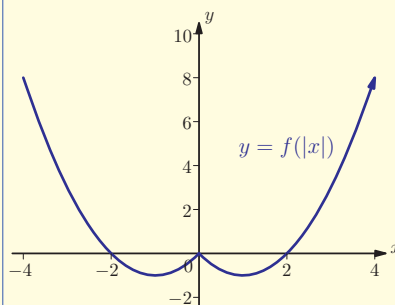
Worked example 6.5

If $f(x) = x(x - 2)$, draw the graph of $y = f(|x|)$.

Draw the graph of $y = f(x)$ for $x \geq 0$



Reflect in the y -axis



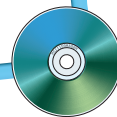
You may have to solve equations involving the modulus function by finding the intersections of two graphs.

KEY POINT 6.9

When solving an equation involving a modulus function, sketch the graph.

EXAM HINT

Your GDC can sketch modulus functions. See Calculator sheet 3 on the CD-ROM.



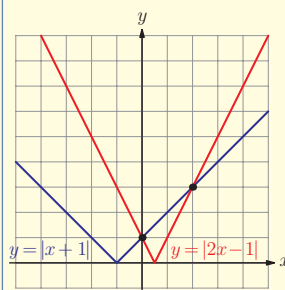
Remember that, when sketching a graph of the form $y = |f(x)|$ you reflect a part of the graph in the x -axis. When finding the coordinates of an intersection point you therefore need to decide whether it is on the reflected or the unreflected part of the graph. If it is on the unreflected part you simply remove the modulus signs and solve the equation. If it is on the reflected part you replace the modulus sign by the minus sign.

Worked example 6.6



Solve the equation $|x + 1| = |2x - 1|$.

Sketch the graphs of $y = |x + 1|$ and $y = |2x - 1|$



Describe the intersection points

There are two intersections – one with the unreflected blue line and the reflected red line:

$$x + 1 = -(2x - 1) \quad (1)$$

and one with the unreflected blue line and the unreflected red line:

$$x + 1 = 2x - 1 \quad (2)$$

Solve the resulting equations

$$(1) \quad x + 1 = -2x + 1$$

$$\Leftrightarrow 3x = 0$$

$$\Leftrightarrow x = 0$$

$$(2) \quad x + 1 = 2x - 1$$

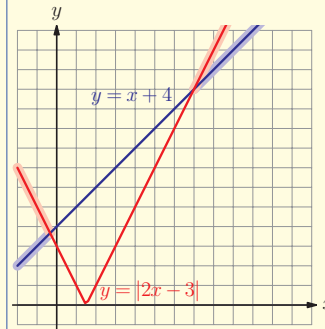
$$\Leftrightarrow x = 2$$

To solve inequalities you also sketch the graphs and find their intersections. You then decide on which parts of the graph the inequality is satisfied.

Worked example 6.7

Solve $|2x - 3| > x + 4$.

Sketch the graphs of $y = |2x - 3|$ and $y = x + 4$, highlighting where the equation is satisfied



Find where the two lines meet

Describe the highlighted region in terms of x

Using GDC:

Lines meet when $x = -\frac{1}{3}$ and when $x = 7$

$x < -\frac{1}{3}$ or $x > 7$

Exercise 6D



1. Sketch the following graphs, showing the axes intercepts.

(a) (i) $y = |2x - 1|$ (ii) $y = |-3x + 2|$

(b) (i) $y = |x| - 2$ (ii) $y = |x| - 2$

(c) (i) $y = |x^2 + x - 2|$ (ii) $y = |x^2 - 5x + 6|$

(d) (i) $y = |x|^2 - 3|x| + 2$ (ii) $y = |x|^2 + |x| - 6$



2. Sketch the following graphs, showing the axes intercepts and any asymptotes.

(a) (i) $y = |x^3 - x^2 - 3|$ (ii) $y = |x^3 + 3x - 1|$

(b) (i) $y = \frac{1}{1 + |x|}$ (ii) $y = \sqrt{1 + 2|x|}$

(c) (i) $y = |\ln(x - 2)|$ (ii) $y = e^{|x-1|}$



3. Solve these equations.

(a) (i) $|x| = 4$ (ii) $|x| = 18$

(b) (i) $|2x - 4| = 4$ (ii) $|3x + 1| = 2$

(c) (i) $|2x + 4| = 4 - x$ (ii) $|5 - 2x| = x + 3$

(d) (i) $|3 - 2x| = |x + 1|$ (ii) $|4 + x| = |5 - 3x|$



4. Solve the following equations.

- (a) (i) $|3x - 4| = 8 - x$ (ii) $|5 + 2x| = 3 - 2x$
 (b) (i) $|6 - x| = |5 + x|$ (ii) $|4 + 3x| = |x|$
 (c) (i) $|x + 1| + |x - 1| = x + 4$ (ii) $|3x - 1| = x + |2 - x|$
 (d) (i) $|x + 1| = x$ (ii) $|x| - 3 = x^2$



5. Solve the following inequalities.

- (a) (i) $|x| > 5$ (ii) $|x| > 2$
 (b) (i) $|x| < 3$ (ii) $|x| < 10$
 (c) (i) $|2x + 1| > 4$ (ii) $|3x - 2| < 3$
 (d) (i) $|2x - 5| < x + 1$ (ii) $|5 - 3x| > 2x$
 (e) (i) $|x + 4| > |2x|$ (ii) $|1 + 3x| < |x + 3|$



6. Solve the inequality $|3x + 1| > 2x$.

[5 marks]



7. Solve the equation $|x^2 - 4| = 8x$.

[5 marks]

8. Solve the equation $|x^2 - 7x + 10| = 7x - 10 - x^2$.

[4 marks]



9. Solve the equation $x|x| = 4x$.

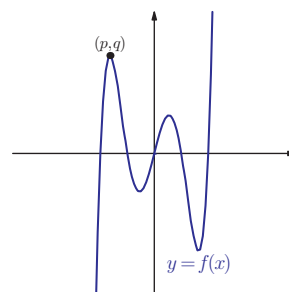
[4 marks]

10. Solve the equation $|x + q^2| = |x - 2q^2|$ giving your answer in terms of q .

[5 marks]

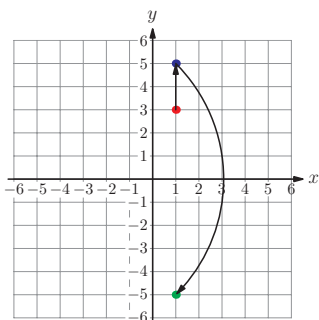
11. Alongside is the graph of $y = f(x)$. Sketch the graph of $y = f(x) + |f(x)|$.

[4 marks]

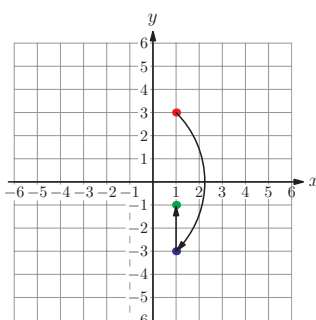


6E Consecutive transformations

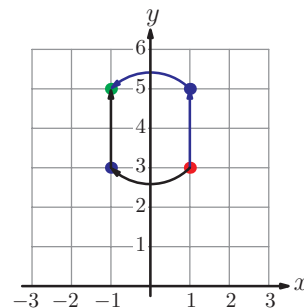
In this section we look at what happens when we apply two transformations in succession.



If the point $(1, 3)$ is translated two units up followed by a reflection in the x -axis (two vertical transformations) the new point is $(1, -5)$.



If $(1, 3)$ is reflected in the x -axis followed by a translation two units up (two vertical transformations) the new point is $(1, -1)$.



However, if the transformations were a translation two units up and a reflection in the y -axis (one vertical transformation and one horizontal transformation), irrespective of the order the result would be $(-1, 5)$.

In advanced mathematics, algebra is much more than using letters to represent numbers. Unknowns can include transformations and many other things.



As you can see in this section, the rules for transformations are different from the rules for numbers, but there are certain similarities too. The study of this more general form of algebra is called group theory, and it is studied further in Option 8: Sets, Relations and Groups. It has many applications from particle physics to painting polyhedra.

KEY POINT 6.10

When two horizontal or two vertical transformations are combined, the outcome depends on the order.

When one vertical and one horizontal transformation are combined, the outcome does not depend on the order.

There is a very important rule to remember when resolving horizontal or vertical transformations:

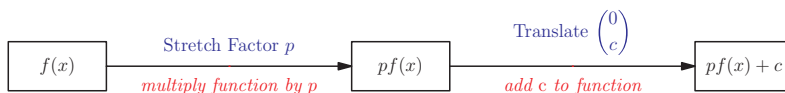
- vertical transformations follow the 'normal' order of operations as applied to arithmetic
- horizontal transformations are resolved in the opposite order to the 'normal' order of operations.

This is demonstrated below:

Let us combine two vertical transformations to transform the graph of $y = f(x)$ into the graph of:

$$y = pf(x) + c$$

We can achieve this by first multiplying $f(x)$ by p and then adding on c , so this form represents a stretch / reflection followed by a translation.

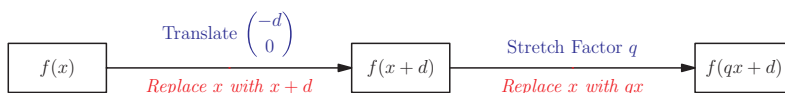


This follows the order of operations as you would expect.

If we combine two horizontal transformations, we can transform the graph of $y = f(x)$ into the graph of:

$$y = f(qx + d)$$

We can achieve this by first replacing x with $x + d$ and then replacing all occurrences of x by qx , so this represents a translation followed by a stretch / reflection.



Following the normal order of operations, you would expect to resolve ' qx ' before ' $+d$ ' but you resolve the transformation in the opposite order.

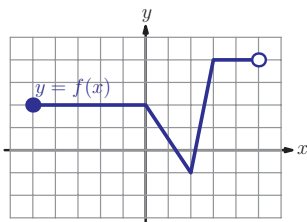
In the first question in Exercise 6E you must consider what happens if you apply these transformations in the reverse order.

EXAM HINT

You may wish to simply remember that when resolving vertical transformations we follow the normal order of operations, while horizontal transformations are resolved in the opposite order.

Worked example 6.8

Below is the graph of $y = f(x)$. Sketch the corresponding graph for $y = 3 - 2f(2x + 1)$.



Break down the transformation into components

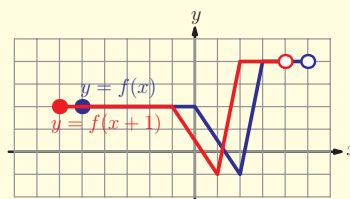
Replace x with $x + 1$
Change $y = f(x)$ to $y = f(x + 1)$

Replace x with $2x$
Change $y = f(x + 1)$ to $y = f(2x + 1)$

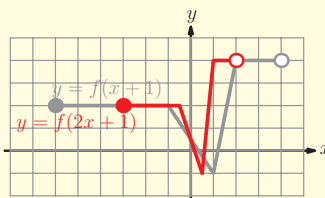
Multiply RHS by -2
Change $y = f(2x + 1)$
to $y = -2f(2x + 1)$

$(2x + 1)$ is two horizontal transformations (changes x)
 $3 - 2f(2x + 1)$ is two vertical transformations (changes y)
Changing x : add 1, then multiply by 2
Changing y : multiply by -2 , then add 3

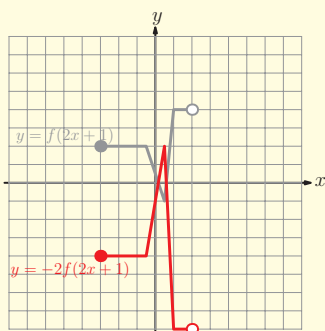
Horizontal Translation $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$



Horizontal Stretch, scale factor $\frac{1}{2}$, relative to $x = 0$



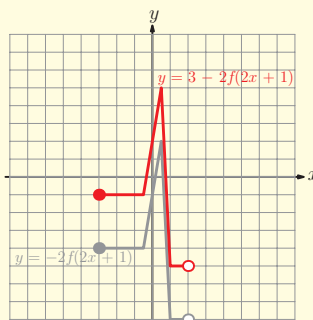
Vertical Reflection through $y = 0$ and
Vertical Stretch, scale factor 2, relative to line $y = 0$



continued...

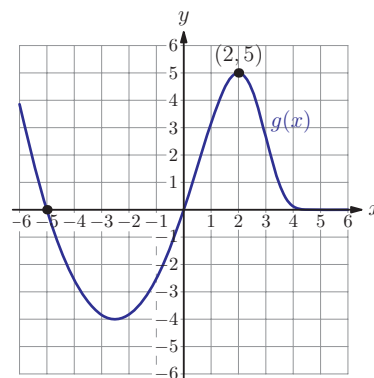
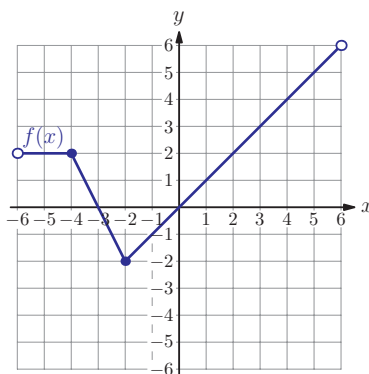
Add 3 to RHS
Change $y = -2f(2x + 1)$
to $y = 3 - 2f(2x + 1)$

Vertical Translation $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$



Exercise 6E

- The graph $y = f(x)$ is transformed by applying first a vertical translation c upwards, then a vertical stretch factor p relative to the x -axis. What is the equation of the resulting graph?
 - The graph $y = f(x)$ is transformed by applying first a horizontal stretch, factor q relative to the y -axis, then a horizontal translation d to the left. What is the equation of the resulting graph?
- Here are the graphs of $y = f(x)$ and $y = g(x)$.



Sketch:

(a) (i) $2f(x) - 1$

(ii) $\frac{1}{2}g(x) + 3$

(b) (i) $4 - f(x)$

(ii) $2 - 2g(x)$

- (c) (i) $3(f(x)-2)$ (ii) $\frac{1-g(x)}{2}$
 (d) (i) $f\left(\frac{x}{2}-1\right)$ (ii) $g(2x+3)$
 (e) (i) $f\left(\frac{4-x}{5}\right)$ (ii) $g\left(\frac{x-3}{2}\right)$

3. For the graphs of $f(x)$ and $g(x)$, sketch

- (a) (i) $|f(x)|$ (ii) $|g(x)|$
 (b) (i) $4+2|f(x)|$ (ii) $2|g(x)|-2$
 (c) (i) $2|2+f(x)|$ (ii) $2|g(x)-1|$

4. For the functions $f(x)$ and $g(x)$ from question 1, sketch

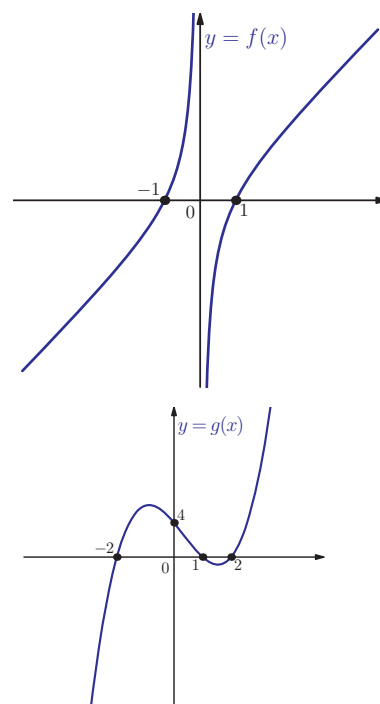
- (a) (i) $f(|x|)$ (ii) $g(|x|)$
 (b) (i) $f(2+|x|)$ (ii) $g(|x|-1)$
 (c) (i) $f(|x-2|+1)$ (ii) $g(|x+1|-1)$

5. If $f(x) = x^2$, express each of the following functions as $af(x)+b$ and hence describe the transformation mapping $f(x)$ to the given function.

- (a) (i) $k(x) = 2x^2 - 6$ (ii) $k(x) = 5x^2 + 4$
 (b) (i) $h(x) = 5 - 3x^2$ (ii) $h(x) = 4 - 8x^2$

6. If $f(x) = 2x^2 - 4$, give the function $g(x)$ which shows the graph of $f(x)$ after:

- (a) (i) translation $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$ followed by a vertical stretch of scale factor 3
 (ii) translation $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$ followed by a vertical stretch of scale factor $\frac{1}{2}$
 (b) (i) vertical stretch of scale factor $\frac{1}{2}$ followed by a translation $\begin{pmatrix} 0 \\ 6 \end{pmatrix}$
 (ii) vertical stretch of scale factor $\frac{7}{2}$ followed by a translation $\begin{pmatrix} 0 \\ 10 \end{pmatrix}$
 (c) (i) reflection through the horizontal axis
 (ii) reflection through the horizontal axis followed by a translation $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$



(d) (i) reflection through the horizontal axis and vertical stretch of scale factor $\frac{1}{2}$ followed by a translation $\begin{pmatrix} 0 \\ 3 \end{pmatrix}$

(ii) reflection through the horizontal axis followed by a translation $\begin{pmatrix} 0 \\ -6 \end{pmatrix}$ followed by a vertical stretch of scale factor $\frac{3}{2}$

7. If $f(x) = x^2$, express each of the following functions as $f(ax + b)$ and hence describe the transformation mapping $f(x)$ to the given function.

(a) (i) $g(x) = x^2 + 2x + 1$ (ii) $g(x) = x^2 - 6x + 9$

(b) (i) $h(x) = 4x^2$ (ii) $h(x) = \frac{x^2}{9}$

(c) (i) $k(x) = 2x^2 + 8x + 4$ (ii) $k(x) = 9x^2 - 6x + 1$

8. If $f(x) = 2x^2 - 4$, give the function g which shows the graph of $f(x)$ after:

(a) (i) translation $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ followed by a horizontal stretch of scale factor $\frac{1}{4}$

(ii) translation $\begin{pmatrix} -2 \\ 0 \end{pmatrix}$ followed by a vertical stretch of scale factor $\frac{1}{2}$

(b) (i) vertical stretch of scale factor $\frac{1}{2}$ followed by a translation $\begin{pmatrix} -4 \\ 0 \end{pmatrix}$

(ii) vertical stretch of scale factor $\frac{2}{3}$ followed by a translation $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

(c) (i) translation $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$ followed by a reflection through the vertical axis

(ii) reflection through the vertical axis followed by a translation $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$

9. For each of the following functions $f(x)$ and $g(x)$, express $g(x)$ in the form $g(x) = af(x+b)+c$, for some values a , b and c , and hence describe a sequence of horizontal and vertical transformations which map $f(x)$ to $g(x)$.

- (a) (i) $f(x) = x^2$ $g(x) = 2x^2 + 4x$
 (ii) $f(x) = x^2$ $g(x) = 3x^2 - 24x + 8$
 (b) (i) $f(x) = x^2 + 3$ $g(x) = -6x + 8$
 (ii) $f(x) = x^2 - 2$ $g(x) = 2 + 8x - 4x^2$

10. (a) The graph of function $f(x) = ax + b$ is transformed by the following sequence:

- translation by $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$
- reflection through $y = 0$
- horizontal stretch, scale factor $\frac{1}{3}$, relative to $x = 0$.

The resulting function is $g(x) = 4 - 15x$. Find a and b .

(b) The graph of function $f(x) = ax^2 + bx + c$ is transformed by the following sequence:

- reflection through $x = 0$
- translation by $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$
- horizontal stretch, scale factor 2, relative to $y = 0$.

The resulting function is $g(x) = 4x^2 + ax - 6$. Find a , b and c .

[10 marks]

11. If $f(x) = 2^x + x$, give in simplest terms the formula for $h(x)$, which is obtained by transforming $f(x)$ by the following sequence of transformations:

- vertical stretch, scale factor 8, relative to $y = 0$
- translation by $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$
- horizontal stretch, scale factor $\frac{1}{2}$, relative to $x = 0$.

[6 marks]



12. Sketch the following graphs:

- (a) $y = \ln x$
 (b) $y = 3\ln(x+2)$
 (c) $y = \ln(2x-1)$

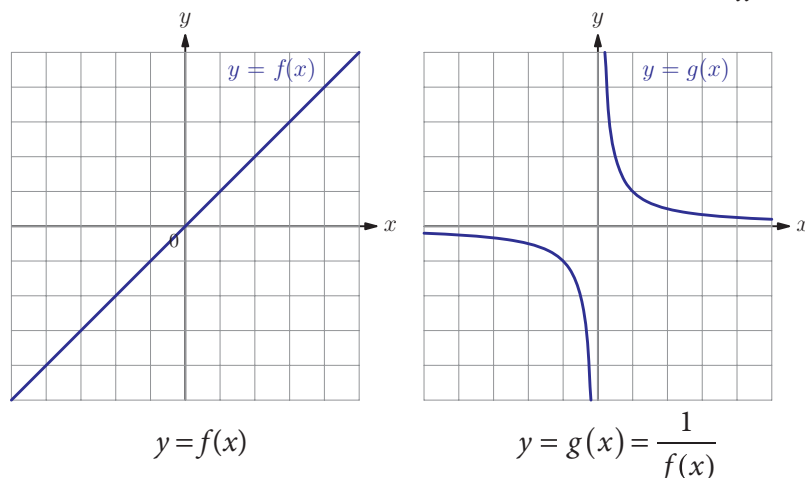
In each case, indicate clearly the positions of the vertical asymptote and the x -intercept.

[6 marks]

6F Reciprocal transformations

A reciprocal transformation turns the graph of $y = f(x)$ into the graph of $y = \frac{1}{f(x)}$.

The simplest reciprocal transformation turns $y = x$ into $y = \frac{1}{x}$.



The table below summarises the relationship between the graphs of $f(x)$ and $g(x) = \frac{1}{f(x)}$

KEY POINT 6.11

$f(x)$	$g(x) = \frac{1}{f(x)}$
$f(a) = 0$	The line $x = a$ is a vertical asymptote
The line $x = a$ is a vertical asymptote	$g(a) = 0$
$f(x) \rightarrow +\infty$	$g(x) \rightarrow 0$ (tends to zero from above)
$f(x) \rightarrow -\infty$	$g(x) \rightarrow 0$ (tends to zero from below)
$f(x) \rightarrow 0$	$g(x) \rightarrow +\infty$ or $-\infty$
$f(x)$ has a horizontal asymptote $y = b$	$g(x)$ has a horizontal asymptote $y = \frac{1}{b}$
$0 < f(x) < 1$	$g(x) > 1$
$f(x) = 1$	$g(x) = 1$
$f(x) > 1$	$0 < g(x) < 1$

$f(x)$	$g(x) = \frac{1}{f(x)}$
$-1 < f(x) < 0$	$g(x) < -1$
$f(x) = -1$	$g(x) = -1$
$f(x) < -1$	$-1 < g(x) < 0$
$(a, f(a))$ is a turning point	$(a, g(a))$ is the opposite turning point, if $g(a)$ is finite

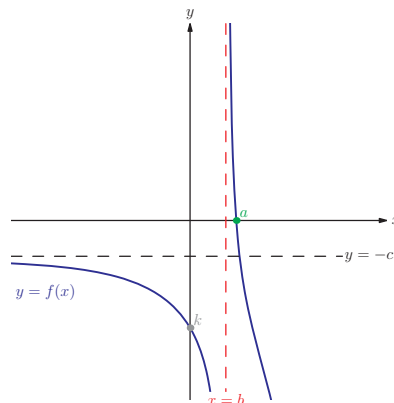
Using these Key points as a guide, it is possible to make a rough sketch of the reciprocal for any given graph.

Worked example 6.9

The diagram shows the graph of an unspecified function

$f(x)$, with a root at a and asymptotes $x = b$ and $y = -c$;

sketch the corresponding graph of $g(x) = \frac{1}{f(x)}$.



What happens when x is large and negative?

As $x \rightarrow -\infty$, $f(x) \rightarrow -c \therefore g(x) \rightarrow \left(-\frac{1}{c}\right)$

So $y = -\frac{1}{c}$ is a horizontal asymptote of $g(x)$

What happens when x is large and positive?

As $x \rightarrow +\infty$, $f(x) \rightarrow -\infty \therefore g(x) \rightarrow 0$

$y = 0$ is a horizontal asymptote of $g(x)$

What happens when $y = f(x)$ has a root?

$f(a) = 0$ so $x = a$ is a vertical asymptote of $g(x)$

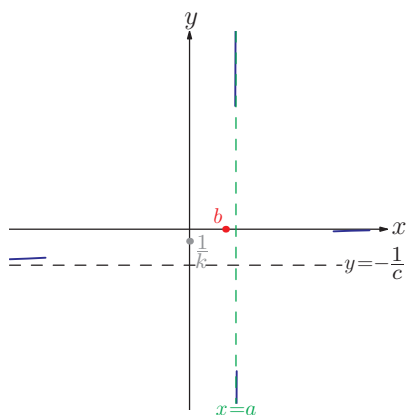
What happens when $y = f(x)$ has a vertical asymptote?

$x = b$ is a vertical asymptote of $f(x)$
so $g(b) = 0$

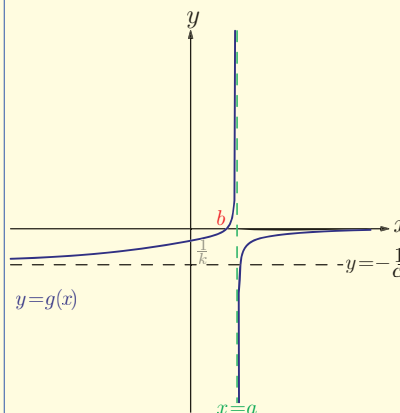
continued . . .

What happens when $y = f(x)$ has a turning point?

Putting this information together:



$f(x)$ has no turning points



EXAM HINT

If you are unsure about which side of an asymptote the graph lies, check a few points.

Exercise 6F



1. By first sketching the graph of $y = f(x)$, sketch the following graphs of the form $y = \frac{1}{f(x)}$, indicating the positions of axes intercepts and asymptotes.

(a) (i) $y = \frac{1}{x-3}$

(ii) $y = \frac{1}{x-5}$

(b) (i) $y = \frac{1}{x+1}$

(ii) $y = \frac{1}{x+3}$

(c) (i) $y = \frac{1}{x^2}$

(ii) $y = \frac{1}{(x-1)^2}$

(d) (i) $y = \frac{1}{x^2-4}$

(ii) $y = \frac{1}{x^2+x-6}$

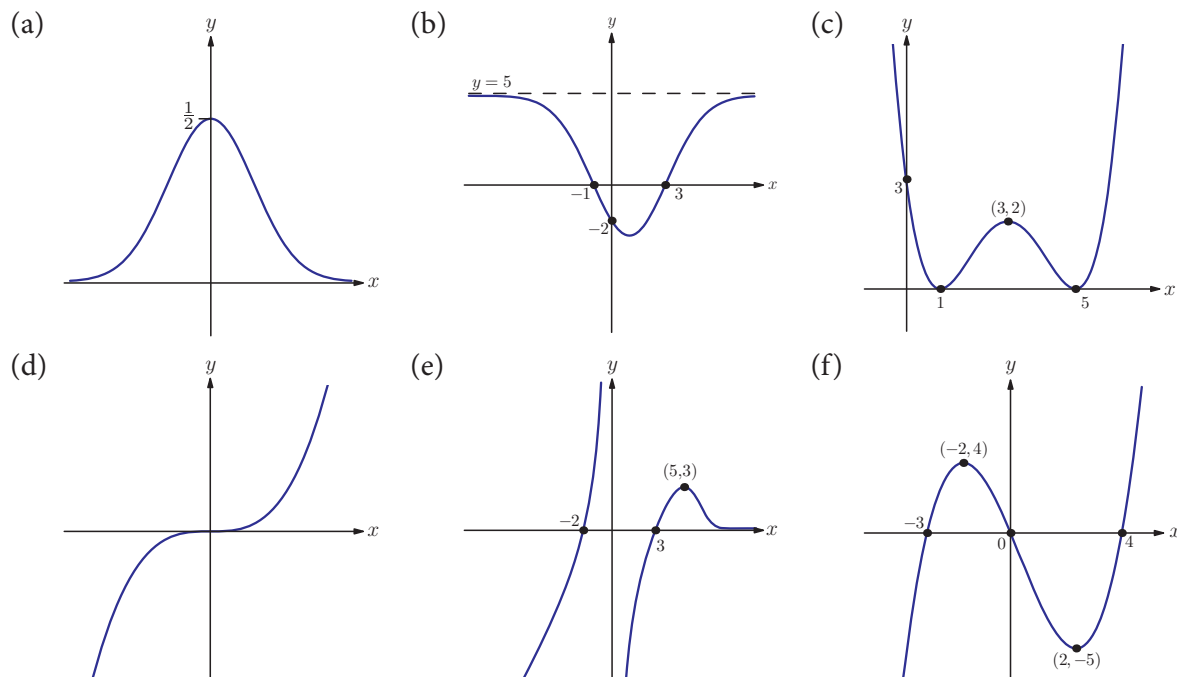
(e) (i) $y = \frac{1}{e^x+1}$

(ii) $y = \frac{1}{e^{x+2}}$

(f) (i) $y = \frac{1}{\ln x+2}$

(ii) $y = \frac{1}{\ln(x+3)}$

2. The graph of $y=f(x)$ is shown. Sketch the graph of $y = \frac{1}{f(x)}$.



3. (a) The graph of $y=f(x)$ is transformed into the graph of $y = f(x-3)$ by a translation with vector $\begin{pmatrix} p \\ q \end{pmatrix}$. State the value of p and the value of q .

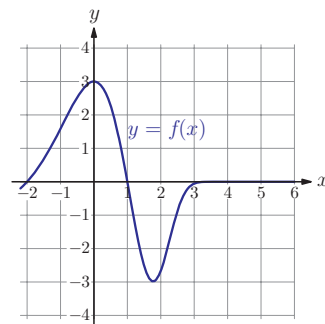
(b) Sketch the following graphs.

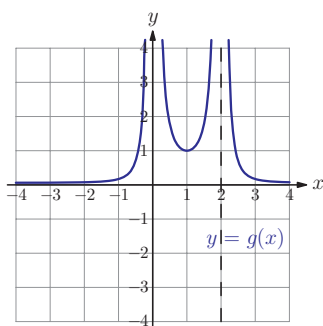
(i) $y = (x-3)^2$ (ii) $y = \frac{1}{(x-3)^2}$. [6 marks]

4. The graph of $y=f(x)$ is shown alongside. Sketch a copy of $y=f(x)$ and on the same diagram, sketch the graph of $y = \frac{1}{f(x)}$,

indicating clearly the positions of any asymptotes and coordinates of any maximum and minimum points.

[6 marks]





5. The graph alongside shows the function $y = g(x)$. On a separate diagram, sketch the graph of

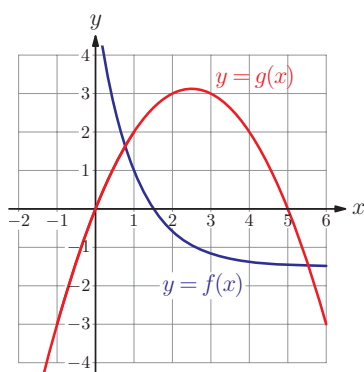
$$y = \frac{1}{g(x)} - 2$$

[6 marks]

6. State the equations of the vertical asymptotes of the graph

$$y = \frac{1}{x^2 e^{x^2} - 4x^2}.$$

[4 marks]



7. The graphs of the functions $y = f(x)$ and $y = g(x)$ are shown alongside.

Sketch a copy of $y = f(x)$ and $y = g(x)$ and, on the same

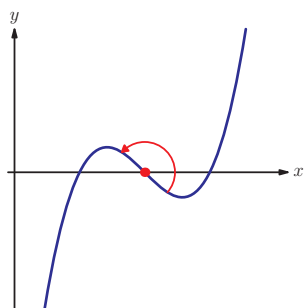
diagram, sketch the graph of $y = \frac{f(x)}{g(x)}$. [5 marks]

6G Symmetries of graphs and functions

There are several possible ways that a graph can be symmetrical. It can have:

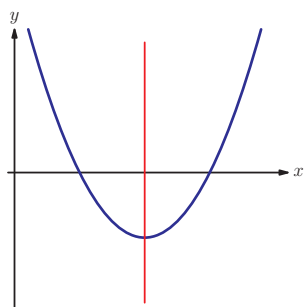
- Two-fold rotational symmetry

When the graph is rotated by a 180° about a given point, it becomes the same graph.



- Reflection symmetry

When the graph is reflected in a given line it becomes the same graph.

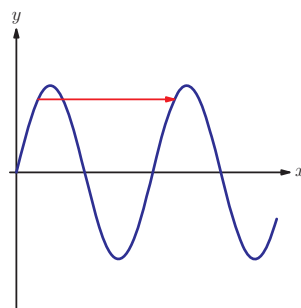
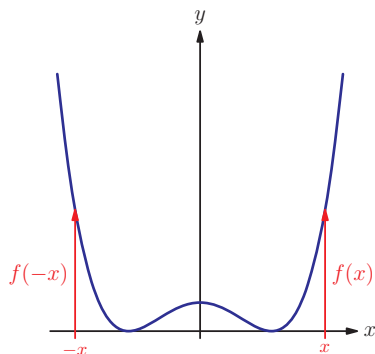


- Translational symmetry

When the graph is shifted it becomes the same graph.

In this section we shall look at how the algebraic properties of the function are related to the symmetry properties of the graph.

When the graph has reflection symmetry in the y -axis, its function is called an **even function**. The height of the graph at every value of x must be the same as the height of the graph at $-x$:



In chapter 10 you will find that trigonometric functions have translational symmetry.

KEY POINT 6.12

For an even function $f(x)$:

$$f(x) = f(-x)$$

This means that $y = f(x)$ has reflection symmetry in the y -axis.



Although quite hard to imagine, there is also a symmetry where the graph becomes the same when it is enlarged. Graphs like this are called fractals.

Worked example 6.10

Consider the function $h(x) = e^x + e^{-x}$.

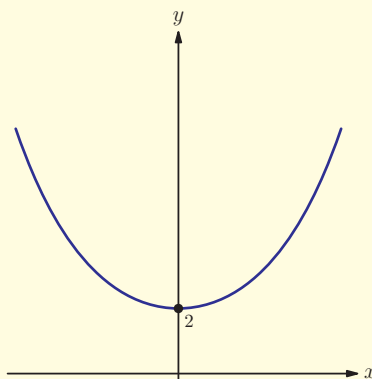
- Find $h(0)$.
- Prove that $h(x)$ is even.
- Sketch the graph of $y = h(x)$.

$$\begin{aligned} \text{(a) } h(0) &= e^0 + e^0 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \text{(b) } h(-x) &= e^{-x} + e^{-(-x)} \\ &= e^{-x} + e^x \\ &= h(x) \end{aligned}$$

continued . . .

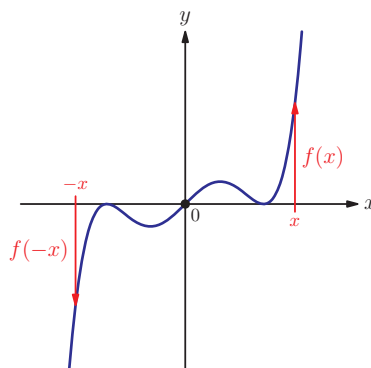
(c) When x is very large $h(x)$ also gets very large, so the graph should look like:



Although the graph above looks like a parabola, it is actually called a *catenary*. A rope hanging between two level points will form this shape.



When the graph has two-fold rotation symmetry about the origin its function is called an **odd function**. The height above the x -axis of the graph at every value of x must be the same as the height below the x -axis of the graph at x , therefore:



KEY POINT 6.13

For an odd function $f(x)$:

$$f(-x) = -f(x)$$

This means that $y = f(x)$ has two-fold rotation symmetry about the origin.

Worked example 6.11

$f(x)$ is an odd function and $g(x)$ is an even function. Prove that $h(x) = f(x)g(x)$ is an odd function.

Describe the symmetries of $f(x)$ and $g(x)$ mathematically

If $h(x)$ is an odd function then we need to show that $h(-x) = -h(x)$. Let's start with $h(-x)$

$$f(-x) = -f(x)$$

$$g(-x) = g(x)$$

$$h(-x) = f(-x)g(-x)$$

$$= -f(x)g(x)$$

$$= -h(x)$$

Therefore $h(x)$ is an odd function.

Exercise 6G

1. Decide if the following functions are odd, even or neither:

- | | |
|---------------------------|--------------------------|
| (a) (i) $y = 3x^4 + 1$ | (ii) $y = x^6 - x^2$ |
| (b) (i) $y = 4x^3 + x$ | (ii) $y = 5x^5$ |
| (c) (i) $y = 1 + x$ | (ii) $y = x^2 - x$ |
| (d) (i) $y = \frac{1}{x}$ | (ii) $y = \frac{1}{x^2}$ |
| (e) (i) $y = e^x$ | (ii) $y = \log x$ |

2. (a) Describe fully the symmetry of the graph of an odd function.

(b) Complete the following graph for negative values of x to make $f(x)$:

- (i) an even function
(ii) an odd function.

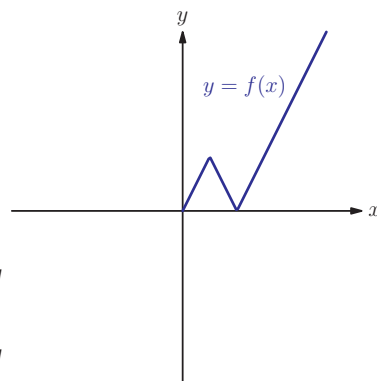
[2 marks]

3. Show that $f(x) = e^{x^2}(1 + x^4)$ is an even function. [3 marks]

4. Explain why an even function can never be a one-to-one function.

5. Prove that the sum of two even functions is an even function.

[4 marks]



6. Prove that the product of two odd functions is an even function. [4 marks]

7. Prove that if $f(x) = ax^n$ is an odd function then n is odd. [4 marks]

8. $f(x)$ is an odd function and $g(x)$ is an even function. Prove that $gf(x)$ is an even function. [4 marks]

9. (a) State the line of symmetry of $f(x) = x^2 + 6x + 7$.
(b) If $f(x - a)$ is an even function find the value of a . [5 marks]

10. The function $f(x)$ satisfies $f(x) = f(10 - x)$. Describe fully the symmetry of the graph of $y = f(x)$. [3 marks]

11. The function $f(x)$ is a one-to-one function over all real numbers. The graph of $y = f(x)$ is symmetrical in the line $y = x$. Evaluate, with justification, $f \circ f(4)$. [3 marks]



12. (a) If $f(x)$ is any function show that $\frac{1}{2}(f(x) - f(-x))$ is an odd function.

(b) Find a similar expression which will always be an even function.

(c) Hence or otherwise prove that all functions can be written as the sum of an odd and an even function.

(d) $f(x) = e^x$ can be written as $g(x) + h(x)$ where $g(x)$ is an odd function and $h(x)$ is an even function.

(i) Sketch $y = g(x)$, labelling all axes intercepts.

(ii) Sketch $y = h(x)$, labelling all axes intercepts. [14 marks]

13. Prove that any odd function which is a polynomial passes through the origin. Give an example to show that this is not necessarily true if the function is not a polynomial.

Summary

- These transformations must be learnt.

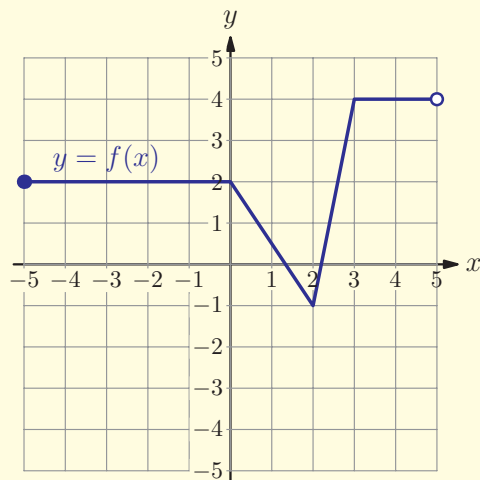
Transformation of $y = f(x)$	Transformation of graph
$y = f(x) + c$	Translation $\begin{pmatrix} 0 \\ c \end{pmatrix}$
$y = f(x + d)$	Translation $\begin{pmatrix} -d \\ 0 \end{pmatrix}$
$y = pf(x)$	Vertical stretch factor p relative to the x -axis: when $p > 0$, stretches away from x -axis when $0 < p < 1$, stretches towards x -axis when $p < 0$, stretch by p and then reflect in x -axis.
$y = f(qx)$	Horizontal stretch factor $\frac{1}{q}$ relative to the y -axis: when $q > 0$, stretches towards y -axis when $0 < q < 1$, stretches away from y -axis when $q < 0$ stretch by $\frac{1}{q}$ then reflect in y -axis
$y = -f(x)$	Reflection in the x -axis
$y = f(-x)$	Reflection in the y -axis
$y = f(x) $	The part of the graph which is below the x -axis is reflected in the x -axis.
$y = f(x)$	The part of the graph which is to the left of the y -axis disappears. The part of the graph which is to the right of the y -axis is reflected in the y -axis.

- The order in which transformations occur may affect the outcome.
- To sketch reciprocal functions we consider the position of zeros, asymptotes and minimum/maximum points on the graph and connect this information together.
- To solve equations and inequalities involving the modulus function, sketch the graph and find intersection points.
- If $f(x)$ is an odd function:
 - $f(-x) = -f(x)$
 - The graph $y = f(x)$ has order two rotational symmetry about the origin.
- If $f(x)$ is an even function:
 - $f(-x) = f(x)$
 - The graph $y = f(x)$ has reflection symmetry in the y -axis.

Introductory problem revisited

Here is the graph of $y = f(x)$.

Sketch the graph of $y = \frac{1}{f(x)}$.



We look for the significant features of the graph.

Where $f(x)$ is constant, $\frac{1}{f(x)}$ is also constant.

So $\frac{1}{f(x)} = \frac{1}{2}$ for $x \leq 0$ and $\frac{1}{f(x)} = \frac{1}{4}$ for $x \geq 3$.

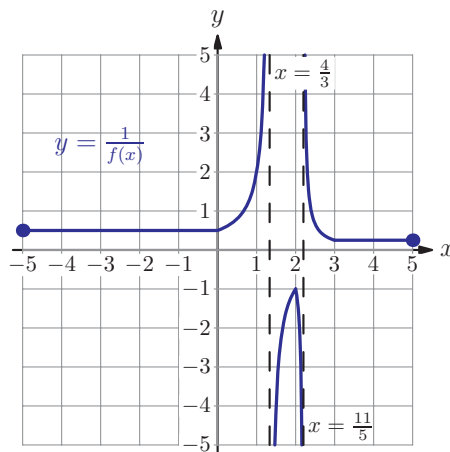
$f(x)$ has two zeros, so $\frac{1}{f(x)}$ has two vertical asymptotes.

$f(x)$ is negative between the zeros and positive on either side, so the same is true for $\frac{1}{f(x)}$.

$f(x)$ has a minimum point with $x = 2$, $y = -1$.

Hence $\frac{1}{f(x)}$ has a maximum point with $x = 2$, $y = \frac{1}{-1} = -1$.

We can now sketch the graph:



Mixed examination practice 6

Short questions

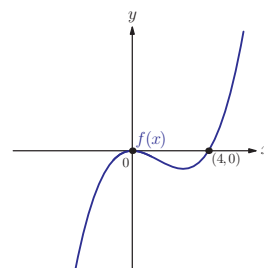
1. The graph of $y = f(x)$ is shown.

Sketch on separate diagrams the graphs of

(a) $y = 3f(x-2)$

(b) $\frac{1}{f(x)}$

Indicate clearly the positions of any x -intercepts and asymptotes. [6 marks]



2. The graph of $y = x^3 - 1$ is transformed by applying a translation with vector $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$ followed by a vertical stretch with scale factor 2. Find the equation of the resulting graph in the form $y = ax^3 + bx^2 + cx + d$. [4 marks]

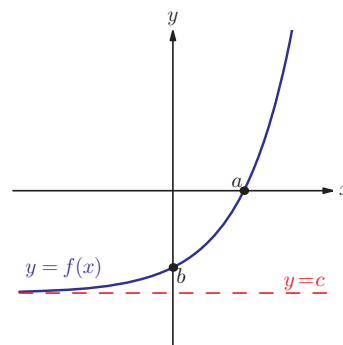
3. Solve the inequality $|2x - 1| < x$. [6 marks]

4. The diagram shows the graph of $y = f(x)$.

On separate diagrams sketch the following graphs, labelling appropriately.

(a) $y = |f(x)|$

(b) $y = f(|x|) - 1$ [5 marks]



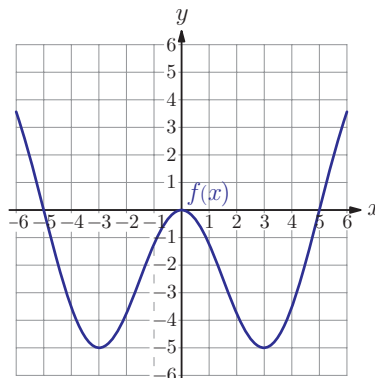
5. (a) Sketch the graph of $y = -\frac{3}{x}$.
 (b) Describe two transformations which transform the graph of $y = \frac{1}{x}$ to the graph of $y = -\frac{3}{x}$.
 (c) Let $f(x) = -\frac{3}{x}, x \neq 0$. Write down an equation for $f^{-1}(x)$. [4 marks]

6. The graph of $y = f(x)$ is shown.

(a) On the same diagram sketch the graph of $y = \frac{1}{f(x)}$.

(b) State the coordinates of the maximum points.

[5 marks]



7. Find two transformations whose composition transforms the graph of $y = (x-1)^2$ to the graph of $y = 3(x+2)^2$. [4 marks]

8. (a) Describe two transformations whose composition transforms the graph of $y = f(x)$ to the graph of $y = 3f\left(\frac{x}{2}\right)$.

(b) Sketch the graph of $y = 3\ln\left(\frac{x}{2}\right)$.

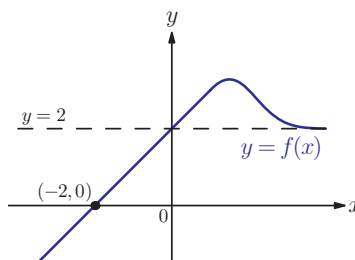
(c) Sketch the graph of $y = 3\ln\left(\frac{x}{2} + 1\right)$ marking clearly the positions of any asymptotes and x -intercepts. [7 marks]

9. The diagram shows a part of the graph of $y = f(x)$. On separate diagrams sketch the graphs of

(a) $y = \frac{1}{f(x)}$

(b) $y = xf(x)$

[6 marks]



10. For which values of the real number x is $|x+k| = |x| + k$, where k is a positive real number?

[4 marks]

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Long questions



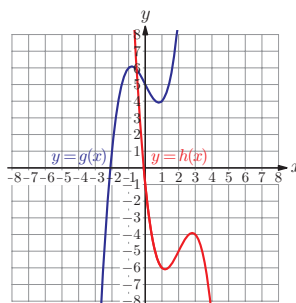
- 1 (a) Describe two transformations which transform the graph of $y = x^2$ to the graph of $y = 3x^2 - 12x + 12$.
- (b) Describe two transformations which transform the graph of $y = x^2 + 6x - 1$ to the graph of $y = x^2$.
- (c) Hence describe a sequence of transformations which transform the graph of $y = x^2 + 6x - 1$ to the graph of $y = 3x^2 - 12x + 12$.
- (d) Sketch the graph of $y = \frac{1}{3x^2 - 12x + 12}$. [12 marks]

2. Given that $f(x) = \frac{3x-5}{x-2}$

- (a) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.
- (b) Find the value of constants p and q such that $f(x) = p + \frac{q}{x-2}$.
- (c) Hence describe a single transformation which transforms the graph of $y = \frac{1}{x}$ to the graph of $y = f(x)$.
- (d) Find an expression for $f^{-1}(x)$ and state its domain.
- (e) Describe the transformation which transforms the graph of $y = f(x)$ to the graph of $y = f^{-1}(x)$. [11 marks]



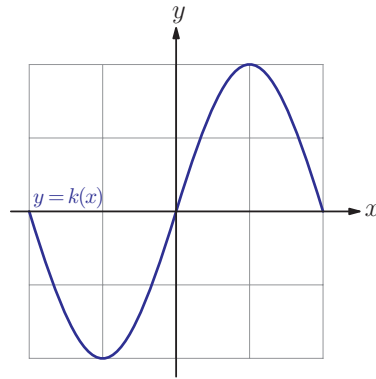
3. (a) Describe a transformation which transforms the graph of $y = f(x)$ to the graph of $y = f(x+2)$.
- (b) Sketch on the same diagram the graphs of
- (i) $y = \ln(x+2)$ (ii) $y = \frac{1}{\ln(x+2)}$.
- Mark clearly any asymptotes and x -intercepts on your sketches.
- (c) The graph of the function $y = g(x)$ has been translated and then reflected in the x -axis to produce the graph of $y = h(x)$.



(i) State the translation vector.

(ii) If $g(x) = x^3 - 2x + 5$, find constants a , b , c and d such that $h(x) = ax^3 + bx^2 + cx + d$.

(d) The diagram below shows the graph of $y = k(x)$.



On the same diagram, sketch the graph of $y = (k(x))^2$.

[14 marks]



4. $f(x) = x^2 - 7x + 10$ $g(x) = x^2 - 7|x| + 10$

(a) Sketch the graph of $y = f(x)$.

(b) Show that $g(x) = f(|x|)$.

(c) Sketch the graph of $y = g(x)$.

(d) Solve the equation $g(x) = x^2$.

(e) Solve the equation $g(x) = -2$.

[12 marks]

5. If $f(x) = 3x^2 + bx + 10$ and the graph $y = f(x)$ has a line of symmetry when $x = 3$

(a) find b .

(b) If $f(x) = f(d - x)$ for all x , find the value of d .

(c) $g(x) = f(x + p) + q$ and $g(x)$ is an even function which passes through the origin. Find p and q .

[14 marks]

(d) Find the set values which satisfy $g(x) = g(|x|)$.



6. (a) Sketch the graph of $y = e^x - 2$, including the coordinates of all axes intercepts.

(b) On separate axes sketch the graphs of

(i) $y = |e^x - 2|$

(ii) $y = e^{|x|} - 2$

(c) Hence solve the equation $e^{|x|} - 2 = |e^x - 2|$.

[10 marks]